gate1993 set-theory&algebra group-theory normal descriptive

Answer key

4.4.5 Group Theory: GATE CSE 1994 | Question: 1.10

Some group (G, o) is known to be abelian. Then, which one of the following is true for G?

A. $g = g^{-1}$ for every $g \in G$

B. $g = g^2$ for every $g \in G$

C. $(goh)^2 = g^2oh^2$ for every $g, h \in G$

D. G is of finite order

gate1994 set-theory&algebra group-theory normal

Answer key

4.4.6 Group Theory: GATE CSE 1995 | Question: 2.17

Let A be the set of all non-singular matrices over real number and let $*$ be the matrix multiplication operation. Then

A. A is closed under $*$ but $\langle A, *\rangle$ is not a semigroup.
B. $\langle A, *\rangle$ is a semigroup but not a monoid.
C. $\langle A, *\rangle$ is a monoid but not a group.
D. $\langle A, *\rangle$ is a a group but not an abelian group.

gate1995 set-theory&algebra group-theory

Answer key

4.4.7 Group Theory: GATE CSE 1995 | Question: 21

Let G_1 and G_2 be subgroups of a group G.

A. Show that $G_1 \cap G_2$ is also a subgroup of G.
B. Is $G_1 \cup G_2$ always a subgroup of G?.

gate1995 set-theory&algebra group-theory normal descriptive proof

Answer key

4.4.8 Group Theory: GATE CSE 1996 | Question: 1.4

Which of the following statements is FALSE?

A. The set of rational numbers is an abelian group under addition
B. The set of integers in an abelian group under addition
C. The set of rational numbers form an abelian group under multiplication
D. The set of real numbers excluding zero is an abelian group under multiplication

gate1996 set-theory&algebra group-theory normal

Answer key

4.4.9 Group Theory: GATE CSE 1996 | Question: 2.4

Which one of the following is false?

A. The set of all bijective functions on a finite set forms a group under function composition
B. The set $\{1, 2, \dots p-1\}$ forms a group under multiplication mod p, where p is a prime number
C. The set of all strings over a finite alphabet forms a group under concatenation
D. A subset $S \neq \emptyset$ of G is a subgroup of the group $\langle G, *\rangle$ if and only if for any pair of elements $a, b \in S, a * b^{-1} \in S$

gate1996 set-theory&algebra normal set-theory group-theory

Answer key

4.4.10 Group Theory: GATE CSE 1997 | Question: 3.1

Let $(Z, *)$ be an algebraic structure where Z is the set of integers and the operation $*$ is defined by $n * m = \max(n, m)$. Which of the following statements is true for $(Z, *)$?

A. $(Z, *)$ is a monoid
B. $(Z, *)$ is an Abelian group
C. $(Z, *)$ is a group
D. None of the above

gate1997 set-theory&algebra group-theory normal

Answer key

4.4.11 Group Theory: GATE CSE 1998 | Question: 12

Let $(A, *)$ be a semigroup, Furthermore, for every a and b in A, if $a \neq b$, then $a * b \neq b * a$.

a. Show that for every a in A, $a * a = a$
b. Show that for every a, b in A, $a * b * a = a$
c. Show that for every a, b, c in A, $a * b * c = a * c$

gate1998 set-theory&algebra group-theory descriptive

Answer key

4.4.12 Group Theory: GATE CSE 1999 | Question: 4

Let G be a finite group and H be a subgroup of G. For $a \in G$, define $aH = \{ah \mid h \in H\}$.

a. Show that $|aH| = |bH|$.
b. Show that for every pair of elements $a, b \in G$, either $aH = bH$ or aH and bH are disjoint.
c. Use the above to argue that the order of H must divide the order of G.

gate1999 set-theory&algebra group-theory descriptive proof

Answer key

4.4.13 Group Theory: GATE CSE 2000 | Question: 4

Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$ and $\otimes$ denote multiplication modulo 8, that is, $x \otimes y = (xy) \mod 8$

a. Prove that $(\{0, 1\}, \otimes)$ is not a group.
b. Write three distinct groups $(G, \otimes)$ where $G \subset S$ and G has 2 elements.

gatecse-2000 set-theory&algebra descriptive group-theory

Answer key

4.4.14 Group Theory: GATE CSE 2002 | Question: 1.6

Which of the following is true?

A. The set of all rational negative numbers forms a group under multiplication.
B. The set of all non-singular matrices forms a group under multiplication.
C. The set of all matrices forms a group under multiplication.
D. Both B and C are true.

gatecse-2002 set-theory&algebra group-theory normal

Answer key

4.4.15 Group Theory: GATE CSE 2003 | Question: 7

Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

A. does not form a group
B. forms a non-commutative group
C. does not have a right identity element
D. forms a group if the empty string is removed from Σ^*

gatecse-2003 set-theory&algebra group-theory normal

Answer key

4.4.16 Group Theory: GATE CSE 2004 | Question: 72

The following is the incomplete operation table of a $4-$element group.

*	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b				
c				

The last row of the table is

A. $c\ a\ e\ b$
B. $c\ b\ a\ e$
C. $c\ b\ e\ a$
D. $c\ e\ a\ b$

gatecse-2004 set-theory&algebra group-theory normal

Answer key

4.4.17 Group Theory: GATE CSE 2005 | Question: 13

The set $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a group under multiplication modulo 15. The inverses of 4 and 7 are respectively:

A. 3 and 13
B. 2 and 11
C. 4 and 13
D. 8 and 14

gatecse-2005 set-theory&algebra normal group-theory

Answer key

4.4.18 Group Theory: GATE CSE 2005 | Question: 46

Consider the set H of all $3 * 3$ matrices of the type

$$\begin{pmatrix} a & f & e \\ 0 & b & d \\ 0 & 0 & c \end{pmatrix}$$

where a, b, c, d, e and f are real numbers and $abc \neq 0.$ Under the matrix multiplication operation, the set H is:

A. a group
B. a monoid but not a group
C. a semi group but not a monoid
D. neither a group nor a semi group

gatecse-2005 set-theory&algebra group-theory normal

Answer key

4.4.19 Group Theory: GATE CSE 2006 | Question: 3

The set $\{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. Given below are four possible reasons. Which one of them is false?

A. It is not closed
B. 2 does not have an inverse
C. 3 does not have an inverse
D. 8 does not have an inverse

gatecse-2006 set-theory&algebra group-theory normal

Answer key

4.4.20 Group Theory: GATE CSE 2007 | Question: 21

How many different non-isomorphic Abelian groups of order 4 are there?

A. 2　B. 3　C. 4　D. 5

gatecse-2007 group-theory normal

Answer key☟

4.4.21 Group Theory: GATE CSE 2009 | Question: 1

Which one of the following is **NOT** necessarily a property of a Group?

A. Commutativity
B. Associativity
C. Existence of inverse for every element
D. Existence of identity

gatecse-2009 set-theory&algebra easy group-theory

Answer key☟

4.4.22 Group Theory: GATE CSE 2009 | Question: 22

For the composition table of a cyclic group shown below:

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

Which one of the following choices is correct?

A. a, b are generators
B. b, c are generators
C. c, d are generators
D. d, a are generators

gatecse-2009 set-theory&algebra normal group-theory

Answer key☟

4.4.23 Group Theory: GATE CSE 2010 | Question: 4

Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms

A. A Group　B. A Ring　C. An integral domain　D. A field

gatecse-2010 set-theory&algebra normal group-theory

Answer key☟

4.4.24 Group Theory: GATE CSE 2014 | Set 3 | Question: 3

Let G be a group with 15 elements. Let L be a subgroup of G. It is known that $L \neq G$ and that the size of L is at least 4. The size of L is __________.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key☟

4.4.25 Group Theory: GATE CSE 2014 | Set 3 | Question: 50

There are two elements x, y in a group $(G, *)$ such that every element in the group can be written as a product of some number of x's and y's in some order. It is known that

$$x * x = y * y = x * y * x * y = y * x * y * x = e$$

where e is the identity element. The maximum number of elements in such a group is ____.

gatecse-2014-set3 set-theory&algebra group-theory numerical-answers normal

Answer key

4.4.26 Group Theory: GATE CSE 2018 | Question: 19

Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____

gatecse-2018 group-theory numerical-answers set-theory&algebra one-mark

Answer key

4.4.27 Group Theory: GATE CSE 2019 | Question: 10

Let G be an arbitrary group. Consider the following relations on G:

- $R_1 : \forall a, b \in G,\ aR_1b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$
- $R_2 : \forall a, b \in G,\ aR_2b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relations?

A. R_1 and R_2
B. R_1 only
C. R_2 only
D. Neither R_1 nor R_2

gatecse-2019 engineering-mathematics discrete-mathematics set-theory&algebra group-theory one-mark

Answer key

4.4.28 Group Theory: GATE CSE 2020 | Question: 18

Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _______.

gatecse-2020 numerical-answers group-theory easy one-mark

Answer key

4.4.29 Group Theory: GATE CSE 2021 | Set 1 | Question: 34

Let G be a group of order 6, and H be a subgroup of G such that $1 < |H| < 6$. Which one of the following options is correct?

A. Both G and H are always cyclic
B. G may not be cyclic, but H is always cyclic
C. G is always cyclic, but H may not be cyclic
D. Both G and H may not be cyclic

gatecse-2021-set1 set-theory&algebra group-theory two-marks

Answer key

4.4.30 Group Theory: GATE CSE 2022 | Question: 17

Which of the following statements is/are TRUE for a group G?

A. If for all $x, y \in G,\ (xy)^2 = x^2y^2$, then G is commutative.
B. If for all $x \in G,\ x^2 = 1$, then G is commutative. Here, 1 is the identity element of G.
C. If the order of G is 2, then G is commutative.
D. If G is commutative, then a subgroup of G need not be commutative.

gatecse-2022 set-theory&algebra group-theory multiple-selects one-mark

Answer key

4.4.31 Group Theory: GATE CSE 2023 | Question: 41

Let X be a set and 2^X denote the powerset of X.

Define a binary operation Δ on 2^X as follows:

$$A \Delta B=(A-B) \cup(B-A).$$

Let $H=\left(2^X, \Delta\right)$. Which of the following statements about H is/are correct?

A. H is a group.
B. Every element in H has an inverse, but H is NOT a group.
C. For every $A \in 2^X$, the inverse of A is the complement of A.
D. For every $A \in 2^X$, the inverse of A is A.

gatecse-2023 set-theory&algebra group-theory multiple-selects two-marks

Answer key☟

4.4.32 Group Theory: GATE CSE 2024 | Set 1 | Question: 42

Consider the operators $\diamond$ and $\square$ defined by $a \diamond b=a+2 b, a \square b=a b$, for positive integers. Which of the following statements is/are TRUE?

A. Operator $\diamond$ obeys the associative law
B. Operator $\square$ obeys the associative law
C. Operator $\diamond$ over the operator $\square$ obeys the distributive law
D. Operator $\square$ over the operator $\diamond$ obeys the distributive law

gatecse-2024-set1 multiple-selects set-theory&algebra group-theory two-marks

Answer key☟

4.4.33 Group Theory: GATE CSE 2024 | Set 2 | Question: 53

Let Z_n be the group of integers $\{0,1,2, \ldots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is ____________.

gatecse-2024-set2 numerical-answers set-theory&algebra group-theory two-marks

Answer key☟

4.5 Identify Function (1)

4.5.1 Identify Function: GATE CSE 2025 | Set 1 | Question: 39

$A=\{0,1,2,3, \ldots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in \mathrm{F}$, we define

$$\left(f_1 \odot f_2\right)(n)=f_1(n)+f_2(n)$$

for every number n in A. Which of the following is/are CORRECT about the mathematical structure $(\mathrm{F}, \odot)$?

A. $(\mathrm{F}, \odot)$ is an Abelian group.
B. $(\mathrm{F}, \odot)$ is an Abelian monoid.
C. $(F, \odot)$ is a non-Abelian group.
D. $(\mathrm{F}, \odot)$ is a non-Abelian monoid.

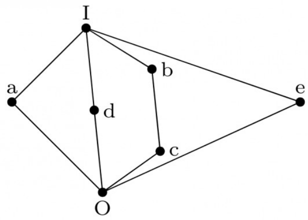
gatecse2025-set1 set-theory&algebra identify-function group-theory multiple-selects easy two-marks

Answer key☟

4.6 Lattice (10)

4.6.1 Lattice: GATE CSE 1988 | Question: 1vii

The complement(s) of the element $'a'$ in the lattice shown in below figure is (are) _____



gate1988 descriptive lattice set-theory&algebra

Answer key☟

4.6.2 Lattice: GATE CSE 1990 | Question: 17c

Show that the elements of the lattice $(N, \leq)$, where N is the set of positive intergers and $a \leq b$ if and only if a divides b, satisfy the distributive property.

gate1990 descriptive set-theory&algebra lattice

Answer key☟

4.6.3 Lattice: GATE CSE 1992 | Question: 14b

Consider the set of integers $\{1, 2, 3, 4, 6, 8, 12, 24\}$ together with the two binary operations LCM (lowest common multiple) and GCD (greatest common divisor). Which of the following algebraic structures does this represent?

A. group
B. ring
C. field
D. lattice

gate1992 set-theory&algebra partial-order lattice normal

Answer key☟

4.6.4 Lattice: GATE CSE 1994 | Question: 2.9

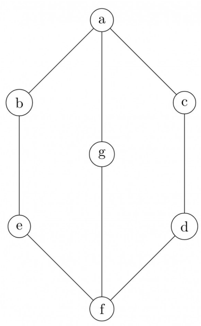
The Hasse diagrams of all the lattices with up to four elements are ________ (write all the relevant Hasse diagrams)

gate1994 set-theory&algebra lattice normal fill-in-the-blanks

Answer key☟

4.6.5 Lattice: GATE CSE 1997 | Question: 3.3

In the lattice defined by the Hasse diagram given in following figure, how many complements does the element 'e' have?



A. 2
B. 3
C. 0
D. 1

gate1997 set-theory&algebra lattice normal

Answer key☟

4.6.6 Lattice: GATE CSE 2002 | Question: 4

$S = \{(1,2),(2,1)\}$ is binary relation on set $A = \{1,2,3\}$. Is it irreflexive? Add the minimum number of ordered pairs to S to make it an equivalence relation. Give the modified S.

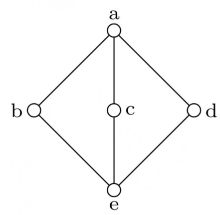
Let $S = \{a, b\}$ and let $\Box(S)$ be the powerset of S. Consider the binary relation '$\subseteq$ (set inclusion)' on $\Box(S)$. Draw the Hasse diagram corresponding to the lattice ($\Box(S), \subseteq$)

gatecse-2002 set-theory&algebra normal lattice descriptive

Answer key☟

4.6.7 Lattice: GATE CSE 2005 | Question: 9

The following is the Hasse diagram of the poset $[\{a, b, c, d, e\}, \prec]$



The poset is :

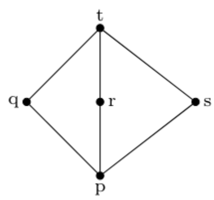
A. not a lattice
B. a lattice but not a distributive lattice
C. a distributive lattice but not a Boolean algebra
D. a Boolean algebra

gatecse-2005 set-theory&algebra lattice normal

Answer key☟

4.6.8 Lattice: GATE CSE 2015 | Set 1 | Question: 34

Suppose $L = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:



For any $x, y \in L$, not necessarily distinct , $x \vee y$ and $x \wedge y$ are join and meet of x, y, respectively. Let $L^3 = \{(x, y, z) : x, y, z \in L\}$ be the set of all ordered triplets of the elements of L. Let p_r be the probability that an element $(x, y, z) \in L^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

A. $p_r = 0$
B. $p_r = 1$
C. $0 < p_r \leq \frac{1}{5}$
D. $\frac{1}{5} < p_r < 1$

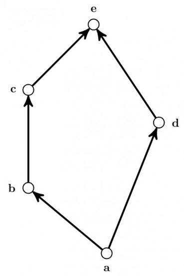
gatecse-2015-set1 set-theory&algebra normal lattice

Answer key☟

4.6.9 Lattice: GATE CSE 2017 | Set 2 | Question: 21

Consider the set $X = \{a, b, c, d, e\}$ under partial ordering $R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}$

The Hasse diagram of the partial order (X, R) is shown below.



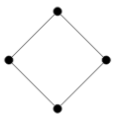
The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is ______

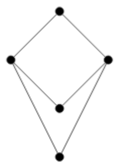
gatecse-2017-set2 set-theory&algebra lattice numerical-answers normal

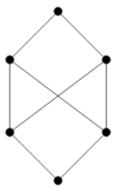
Answer key

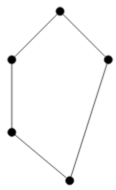
4.6.10 Lattice: GATE IT 2008 | Question: 28

Consider the following Hasse diagrams.


i.


ii.


iii.


iv.

Which all of the above represent a lattice?

A. (i) and (iv) only
B. (ii) and (iii) only
C. (iii) only
D. (i), (ii) and (iv) only

gateit-2008 set-theory&algebra lattice normal

Answer key

4.7 Mathematical Induction (2)

4.7.1 Mathematical Induction: GATE CSE 1995 | Question: 23

Prove using mathematical induction for $n \geq 5, 2^n > n^2$

gate1995 set-theory&algebra proof mathematical-induction descriptive

Answer key

4.7.2 Mathematical Induction: GATE CSE 2000 | Question: 3

Consider the following sequence:

$s_1 = s_2 = 1$ and $s_i = 1 + \min\left(s_{i-1}, s_{i-2}\right)$ for $i > 2$.

Prove by induction on n that $s_n = \lceil \frac{n}{2} \rceil$.

gatecse-2000 set-theory&algebra mathematical-induction descriptive

Answer key

4.8 Number Theory (7)

4.8.1 Number Theory: GATE CSE 1995 | Question: 7(A)

Determine the number of divisors of 600.

gate1995 set-theory&algebra number-theory numerical-answers

Answer key

4.8.2 Number Theory: GATE CSE 2014 | Set 2 | Question: 49

The number of distinct positive integral factors of 2014 is _____________

gatecse-2014-set2 set-theory&algebra easy numerical-answers number-theory

Answer key

4.8.3 Number Theory: GATE CSE 2015 | Set 2 | Question: 9

The number of divisors of 2100 is ____.

gatecse-2015-set2 set-theory&algebra number-theory easy numerical-answers

Answer key

4.8.4 Number Theory: GATE IT 2005 | Question: 34

Let $n = p^2q$, where p and q are distinct prime numbers. How many numbers m satisfy $1 \leq m \leq n$ and gcd $(m, n) = 1$? Note that gcd (m, n) is the greatest common divisor of m and n.

A. $p(q-1)$
B. pq
C. $(p^2-1)(q-1)$
D. $p(p-1)(q-1)$

gateit-2005 set-theory&algebra normal number-theory

Answer key

4.8.5 Number Theory: GATE IT 2007 | Question: 16

The minimum positive integer p such that $3^p \pmod{17} = 1$ is

A. 5
B. 8
C. 12
D. 16

gateit-2007 set-theory&algebra normal number-theory

Answer key

4.8.6 Number Theory: GATE IT 2008 | Question: 24

The exponent of 11 in the prime factorization of $300!$ is

A. 27
B. 28
C. 29
D. 30

gateit-2008 set-theory&algebra normal number-theory

Answer key

4.8.7 Number Theory: GATE1991-15,a

Show that the product of the least common multiple and the greatest common divisor of two positive integers a and b is $a \times b$.

gate1991 set-theory&algebra normal number-theory proof descriptive

Answer key

4.9 Onto (1)

4.9.1 Onto: GATE CSE 2025 | Set 1 | Question: 7

$g(\cdot)$ is a function from A to B, $f(\cdot)$ is a function from B to C, and their composition defined as $f(g(\cdot))$ is a mapping from A to C.

If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$?

A. $g(\cdot)$ must be an onto (surjective) function.
B. $g(\cdot)$ must be a one-to-one (injective) function.
C. $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
D. $g(\cdot)$ is not required to be a one-to-one or onto function.

gatecse2025-set1 set-theory&algebra onto functions one-mark

Answer key

4.10 Partial Order (10)

4.10.1 Partial Order: GATE CSE 1991 | Question: 01,xiv

If the longest chain in a partial order is of length n, then the partial order can be written as a _____ of n antichains.

gate1991 set-theory&algebra partial-order normal fill-in-the-blanks

Answer key

4.10.2 Partial Order: GATE CSE 1993 | Question: 8.5

The less-than relation, $<$, on real number is

A. a partial ordering since it is asymmetric and reflexive
B. a partial ordering since it is antisymmetric and reflexive
C. not a partial ordering because it is not asymmetric and not reflexive
D. not a partial ordering because it is not antisymmetric and reflexive
E. none of the above

gate1993 set-theory&algebra partial-order easy

Answer key

4.10.3 Partial Order: GATE CSE 1996 | Question: 1.2

Let $X = \{2, 3, 6, 12, 24\}$, Let $\leq$ be the partial order defined by $X \leq Y$ if x divides y. Number of edges in the Hasse diagram of $(X, \leq)$ is

A. 3
B. 4
C. 9
D. None of the above

gate1996 set-theory&algebra partial-order normal

Answer key

4.10.4 Partial Order: GATE CSE 1997 | Question: 6.1

A partial order $\leq$ is defined on the set $S = \{x, a_1, a_2, \dots, a_n, y\}$ as $x \leq_i a_i$ for all i and $a_i \leq y$ for all i, where $n \geq 1$. The number of total orders on the set S which contain the partial order $\leq$ is

A. $n!$
B. $n+2$
C. n
D. 1

gate1997 set-theory&algebra partial-order normal

Answer key

4.10.5 Partial Order: GATE CSE 1998 | Question: 11

Suppose $A = \{a, b, c, d\}$ and Π_1 is the following partition of A

$\Pi_1 = \{\{a, b, c\}\ \{d\}\}$

a. List the ordered pairs of the equivalence relations induced by Π_1.
b. Draw the graph of the above equivalence relation.
c. Let $\Pi_2 = \{\{a\}, \{b\}, \{c\}, \{d\}\}$
 $\Pi_3 = \{\{a, b, c, d\}\}$
 and $\Pi_4 = \{\{a, b\}, \{c, d\}\}$
 Draw a Poset diagram of the poset, $\langle \{\Pi_1, \Pi_2, \Pi_3, \Pi_4\}, \text{ refines } \rangle$.

gate1998 set-theory&algebra normal partial-order descriptive

Answer key

4.10.6 Partial Order: GATE CSE 2003 | Question: 31

Let $(S, \leq)$ be a partial order with two minimal elements a and b, and a maximum element c. Let P: S$\rightarrow$ {**True, False**} be a predicate defined on S. Suppose that P(a) = **True**, P(b) = **False** and P(x) $\implies$ P(y) for all $x, y \in S$ satisfying $x \leq y$, where $\implies$ stands for logical implication. Which of the following statements CANNOT be true?

A. P(x) = **True** for all x ∈ S such that x ≠ b
B. P(x) = **False** for all x ∈ S such that x ≠ a and x ≠ c
C. P(x) = **False** for all x ∈ S such that b ≤ x and x ≠ c
D. P(x) = **False** for all x ∈ S such that a ≤ x and b ≤ x

gatecse-2003 set-theory&algebra partial-order normal propositional-logic

Answer key☟

4.10.7 Partial Order: GATE CSE 2004 | Question: 73

The inclusion of which of the following sets into

$S = \{\{1,2\}, \{1,2,3\}, \{1,3,5\}, \{1,2,4\}, \{1,2,3,4,5\}\}$

is necessary and sufficient to make S a complete lattice under the partial order defined by set containment?

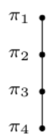
A. $\{1\}$
B. $\{1\},\{2,3\}$
C. $\{1\},\{1,3\}$
D. $\{1\},\{1,3\},\{1,2,3,4\},\{1,2,3,5\}$

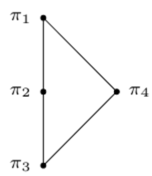
gatecse-2004 set-theory&algebra partial-order normal

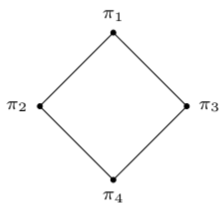
Answer key☟

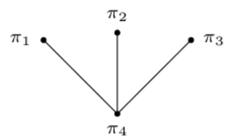
4.10.8 Partial Order: GATE CSE 2007 | Question: 26

Consider the set $S = \{a, b, c, d\}$. Consider the following 4 partitions $\pi_1, \pi_2, \pi_3, \pi_4$ on
$S: \pi_1 = \{\overline{abcd}\}, \quad \pi_2 = \{\overline{ab}, \overline{cd}\}, \quad \pi_3 = \{\overline{abc}, \overline{d}\}, \quad \pi_4 = \{\bar{a}, \bar{b}, \bar{c}, \bar{d}\}$.
Let $\prec$ be the partial order on the set of partitions $S' = \{\pi_1, \pi_2, \pi_3, \pi_4\}$ defined as follows: $\pi_i \prec \pi_j$ if and only if π_i refines π_j. The poset diagram for $(S', \prec)$ is:

A. 

B. 

C. 

D. 

gatecse-2007 set-theory&algebra normal partial-order descriptive

Answer key☟

4.10.9 Partial Order: GATE CSE 2024 | Set 2 | Question: 24

Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$$P = \{(x, x) \mid x \in \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$$

The number of total orders on $\{1,2,3,4\}$ that contain P is __________.

gatecse-2024-set2 numerical-answers set-theory&algebra partial-order one-mark

Answer key☟

4.10.10 Partial Order: GATE IT 2007 | Question: 23

A partial order P is defined on the set of natural numbers as follows. Here $\frac{x}{y}$ denotes integer division.

i. $(0,0) \in P.$
ii. $(a,b) \in P$ if and only if $(a\%10) \leq (b\%10)$ and $\left(\frac{a}{10}, \frac{b}{10}\right) \in P.$

Consider the following ordered pairs:

i. $(101, 22)$
ii. $(22, 101)$
iii. $(145, 265)$
iv. $(0, 153)$

Which of these ordered pairs of natural numbers are contained in P?

A. (i) and (iii)
B. (ii) and (iv)
C. (i) and (iv)
D. (iii) and (iv)

gateit-2007 set-theory&algebra partial-order normal

Answer key☟

4.11 Polynomials (4)

4.11.1 Polynomials: GATE CSE 1997 | Question: 4.4

A polynomial $p(x)$ is such that $p(0) = 5, p(1) = 4, p(2) = 9$ and $p(3) = 20$. The minimum degree it should have is

A. 1
B. 2
C. 3
D. 4

gate1997 set-theory&algebra normal polynomials

Answer key☟

4.11.2 Polynomials: GATE CSE 2000 | Question: 2.4

A polynomial $p(x)$ satisfies the following:

- $p(1) = p(3) = p(5) = 1$
- $p(2) = p(4) = -1$

The minimum degree of such a polynomial is

A. 1
B. 2
C. 3
D. 4

gatecse-2000 set-theory&algebra normal polynomials

Answer key☟

4.11.3 Polynomials: GATE CSE 2014 | Set 2 | Question: 5

A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1$, $x = 2$ and $x = 3$. Which one of the following must be TRUE?

A. $f(0)f(4) < 0$
B. $f(0)f(4) > 0$
C. $f(0) + f(4) > 0$
D. $f(0) + f(4) < 0$

gatecse-2014-set2 set-theory&algebra polynomials normal

Answer key☟

4.11.4 Polynomials: GATE CSE 2017 | Set 2 | Question: 24

Consider the quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b. The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____

gatecse-2017-set2 polynomials numerical-answers set-theory&algebra

Answer key☟

4.12 Relations (38)

4.12.1 Relations: GATE CSE 1987 | Question: 2d

State whether the following statements are TRUE or FALSE:

The union of two equivalence relations is also an equivalence relation.

gate1987 set-theory&algebra relations true-false

Answer key☟

4.12.2 Relations: GATE CSE 1987 | Question: 9a

How many binary relations are there on a set A with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key☟

4.12.3 Relations: GATE CSE 1987 | Question: 9e

How many true inclusion relations are there of the form $A \subseteq B$, where A and B are subsets of a set S with n elements?

gate1987 set-theory&algebra relations descriptive

Answer key☟

4.12.4 Relations: GATE CSE 1989 | Question: 1-iv

The transitive closure of the relation $\{(1,2),(2,3),(3,4),(5,4)\}$ on the set $\{1,2,3,4,5\}$ is ___________.

gate1989 set-theory&algebra relations descriptive

Answer key☟

4.12.5 Relations: GATE CSE 1992 | Question: 15.b

Let S be the set of all integers and let $n > 1$ be a fixed integer. Define for $a, b \in S, aRb$ iff $a - b$ is a multiple of n. Show that R is an equivalence relation and find its equivalence classes for $n = 5$.

gate1992 set-theory&algebra normal relations descriptive

Answer key☟

4.12.6 Relations: GATE CSE 1994 | Question: 2.3

Amongst the properties $\{\text{reflexivity, symmetry, anti-symmetry, transitivity}\}$ the relation $R = \{(x, y) \in N^2 | x \neq y\}$ satisfies _________

gate1994 set-theory&algebra normal relations fill-in-the-blanks

Answer key☟

4.12.7 Relations: GATE CSE 1995 | Question: 1.19

Let R be a symmetric and transitive relation on a set A. Then

A. R is reflexive and hence an equivalence relation
B. R is reflexive and hence a partial order
C. R is reflexive and hence not an equivalence relation
D. None of the above

gate1995 set-theory&algebra relations normal

Answer key☟

4.12.8 Relations: GATE CSE 1996 | Question: 2.2

Let R be a non-empty relation on a collection of sets defined by ${}_{A}R_{B}$ if and only if $A \cap B = \phi$. Then, (pick the true statement)

A. A is reflexive and transitive
B. R is symmetric and not transitive
C. R is an equivalence relation
D. R is not reflexive and not symmetric

gate1996 set-theory&algebra relations normal

Answer key☟

4.12.9 Relations: GATE CSE 1996 | Question: 8

Let F be the collection of all functions $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$. If f and $g \in F$, define an equivalence relation $\sim$ by $f \sim g$ if and only if $f(3) = g(3)$.

a. Find the number of equivalence classes defined by $\sim$.
b. Find the number of elements in each equivalence class.

gate1996 set-theory&algebra relations functions normal descriptive

Answer key☟

4.12.10 Relations: GATE CSE 1997 | Question: 14

Let R be a reflexive and transitive relation on a set A. Define a new relation E on A as

$E = \{(a, b) \mid (a, b) \in R \text{ and } (b, a) \in R\}$

Prove that E is an equivalence relation on A.

Define a relation $\leq$ on the equivalence classes of E as $E_1 \leq E_2$ if $\exists a, b$ such that $a \in E_1, b \in E_2$ and $(a, b) \in R$. Prove that $\leq$ is a partial order.

gate1997 set-theory&algebra relations normal proof descriptive

Answer key☟

4.12.11 Relations: GATE CSE 1997 | Question: 6.3

The number of equivalence relations of the set $\{1, 2, 3, 4\}$ is

A. 15
B. 16
C. 24
D. 4

gate1997 set-theory&algebra relations normal group-theory

Answer key☟

4.12.12 Relations: GATE CSE 1998 | Question: 1.6

Suppose A is a finite set with n elements. The number of elements in the largest equivalence relation of A is

A. n
B. n^2
C. 1
D. $n + 1$

gate1998 set-theory&algebra relations easy

Answer key☟

4.12.13 Relations: GATE CSE 1998 | Question: 1.7

Let R_1 and R_2 be two equivalence relations on a set. Consider the following assertions:

i. $R_1 \cup R_2$ is an equivalence relation
ii. $R_1 \cap R_2$ is an equivalence relation

Which of the following is correct?

A. Both assertions are true
B. Assertions (i) is true but assertions (ii) is not true
C. Assertions (ii) is true but assertions (i) is not true
D. Neither (i) nor (ii) is true

gate1998 set-theory&algebra relations normal

Answer key

4.12.14 Relations: GATE CSE 1998 | Question: 10a

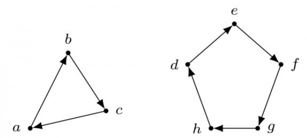
Prove by induction that the expression for the number of diagonals in a polygon of n sides is $\frac{n(n-3)}{2}$

gate1998 set-theory&algebra descriptive relations

Answer key

4.12.15 Relations: GATE CSE 1998 | Question: 10b

Let R be a binary relation on $A = \{a, b, c, d, e, f, g, h\}$ represented by the following two component digraph. Find the smallest integers m and n such that $m < n$ and $R^m = R^n$.



gate1998 descriptive set-theory&algebra relations

Answer key

4.12.16 Relations: GATE CSE 1998 | Question: 2.3

The binary relation $R = \{(1,1), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4)\}$ on the set $A = \{1, 2, 3, 4\}$ is

A. reflexive, symmetric and transitive
B. neither reflexive, nor irreflexive but transitive
C. irreflexive, symmetric and transitive
D. irreflexive and antisymmetric

gate1998 set-theory&algebra easy relations

Answer key

4.12.17 Relations: GATE CSE 1999 | Question: 1.2

The number of binary relations on a set with n elements is:

A. n^2
B. 2^n
C. 2^{n^2}
D. None of the above

gate1999 set-theory&algebra relations combinatory easy

Answer key

4.12.18 Relations: GATE CSE 1999 | Question: 2.3

Let L be a set with a relation R which is transitive, anti-symmetric and reflexive and for any two elements $a, b \in L$, let the least upper bound $lub(a, b)$ and the greatest lower bound $glb(a, b)$ exist. Which of the following is/are true?

A. L is a poset
B. L is a Boolean algebra
C. L is a lattice
D. None of the above

gate1999 set-theory&algebra normal relations multiple-selects

Answer key

4.12.19 Relations: GATE CSE 1999 | Question: 3

a. Mr. X claims the following:
   If a relation R is both symmetric and transitive, then R is reflexive. For this, Mr. X offers the following proof:
   "From xRy, using symmetry we get yRx. Now because R is transitive xRy and yRx together imply xRx. Therefore, R is reflexive".
b. Give an example of a relation R which is symmetric and transitive but not reflexive.

gate1999 set-theory&algebra relations normal descriptive

Answer key

4.12.20 Relations: GATE CSE 2000 | Question: 2.5

A relation R is defined on the set of integers as xRy iff $(x+y)$ is even. Which of the following statements is true?

A. R is not an equivalence relation
B. R is an equivalence relation having 1 equivalence class
C. R is an equivalence relation having 2 equivalence classes
D. R is an equivalence relation having 3 equivalence classes

gatecse-2000 set-theory&algebra relations normal

Answer key

4.12.21 Relations: GATE CSE 2001 | Question: 1.2

Consider the following relations:

- $R_1\ (a, b)$ iff $(a+b)$ is even over the set of integers
- $R_2\ (a, b)$ iff $(a+b)$ is odd over the set of integers
- $R_3\ (a, b)$ iff $a.\,b > 0$ over the set of non-zero rational numbers
- $R_4\ (a, b)$ iff $|a-b| \leq 2$ over the set of natural numbers

Which of the following statements is correct?

A. R_1 and R_2 are equivalence relations, R_3 and R_4 are not
B. R_1 and R_3 are equivalence relations, R_2 and R_4 are not
C. R_1 and R_4 are equivalence relations, R_2 and R_3 are not
D. R_1, R_2, R_3 and R_4 all are equivalence relations

gatecse-2001 set-theory&algebra normal relations

Answer key

4.12.22 Relations: GATE CSE 2002 | Question: 2.17

The binary relation $S = \phi(\text{empty set})$ on a set $A = \{1, 2, 3\}$ is

A. Neither reflexive nor symmetric
B. Symmetric and reflexive
C. Transitive and reflexive
D. Transitive and symmetric

gatecse-2002 set-theory&algebra normal relations

Answer key

4.12.23 Relations: GATE CSE 2002 | Question: 3

Let A be a set of $n(>0)$ elements. Let N_r be the number of binary relations on A and let N_f be the number of functions from A to A

A. Give the expression for N_r, in terms of n.
B. Give the expression for N_f, terms of n.

C. Which is larger for all possible n, N_r or N_f

gatecse-2002 set-theory&algebra normal descriptive relations

Answer key

4.12.24 Relations: GATE CSE 2004 | Question: 24

Consider the binary relation:

$S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2\}\}$

The reflexive transitive closure is S is

A. $\{(x, y) \mid y > x \text{ and } x, y \in \{0, 1, 2\}\}$
B. $\{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2\}\}$
C. $\{(x, y) \mid y < x \text{ and } x, y \in \{0, 1, 2\}\}$
D. $\{(x, y) \mid y \leq x \text{ and } x, y \in \{0, 1, 2\}\}$

gatecse-2004 set-theory&algebra easy relations

Answer key

4.12.25 Relations: GATE CSE 2005 | Question: 42

Let R and S be any two equivalence relations on a non-empty set A. Which one of the following statements is TRUE?

A. $R \cup S$, $R \cap S$ are both equivalence relations
B. $R \cup S$ is an equivalence relation
C. $R \cap S$ is an equivalence relation
D. Neither $R \cup S$ nor $R \cap S$ are equivalence relations

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.26 Relations: GATE CSE 2005 | Question: 7

The time complexity of computing the transitive closure of a binary relation on a set of n elements is known to be:

A. $O(n)$ B. $O(n \log n)$ C. $O\left(n^{\frac{3}{2}}\right)$ D. $O(n^3)$

gatecse-2005 set-theory&algebra normal relations

Answer key

4.12.27 Relations: GATE CSE 2006 | Question: 4

A relation R is defined on ordered pairs of integers as follows:

$$(x, y) R(u, v) \text{ if } x < u \text{ and } y > v$$

Then R is:

A. Neither a Partial Order nor an Equivalence Relation
B. A Partial Order but not a Total Order
C. A total Order
D. An Equivalence Relation

gatecse-2006 set-theory&algebra normal relations

Answer key

4.12.28 Relations: GATE CSE 2007 | Question: 2

Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are:

A. n and n
B. n^2 and n
C. n^2 and 0
D. n and 1

gatecse-2007 set-theory&algebra normal relations

Answer key☟

4.12.29 Relations: GATE CSE 2009 | Question: 4

Consider the binary relation $R = \{(x,y),(x,z),(z,x),(z,y)\}$ on the set $\{x,y,z\}$. Which one of the following is **TRUE**?

A. R is symmetric but NOT antisymmetric
B. R is NOT symmetric but antisymmetric
C. R is both symmetric and antisymmetric
D. R is neither symmetric nor antisymmetric

gatecse-2009 set-theory&algebra easy relations

Answer key☟

4.12.30 Relations: GATE CSE 2010 | Question: 3

What is the possible number of reflexive relations on a set of 5 elements?

A. 2^{10}
B. 2^{15}
C. 2^{20}
D. 2^{25}

gatecse-2010 set-theory&algebra easy relations

Answer key☟

4.12.31 Relations: GATE CSE 2015 | Set 2 | Question: 16

Let R be the relation on the set of positive integers such that aRb and only if a and b are distinct and let have a common divisor other than 1. Which one of the following statements about R is true?

A. R is symmetric and reflexive but not transitive
B. R is reflexive but not symmetric not transitive
C. R is transitive but not reflexive and not symmetric
D. R is symmetric but not reflexive and not transitive

gatecse-2015-set2 set-theory&algebra relations normal

Answer key☟

4.12.32 Relations: GATE CSE 2015 | Set 3 | Question: 41

Let R be a relation on the set of ordered pairs of positive integers such that $((p,q),(r,s)) \in R$ if and only if $p-s=q-r$. Which one of the following is true about R?

A. Both reflexive and symmetric
B. Reflexive but not symmetric
C. Not reflexive but symmetric
D. Neither reflexive nor symmetric

gatecse-2015-set3 set-theory&algebra relations normal

Answer key☟

4.12.33 Relations: GATE CSE 2016 | Set 2 | Question: 26

A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a,b)R(c,d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

- $P : R$ is reflexive.
- $Q : R$ is transitive.

Which one of the following statements is **TRUE**?

A. Both P and Q are true.
B. P is true and Q is false.
C. P is false and Q is true.
D. Both P and Q are false.

gatecse-2016-set2 set-theory&algebra relations normal

Answer key

4.12.34 Relations: GATE CSE 2020 | Question: 17

Let $\mathcal{R}$ be the set of all binary relations on the set $\{1,2,3\}$. Suppose a relation is chosen from $\mathcal{R}$ at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is ______.

gatecse-2020 numerical-answers probability relations one-mark

Answer key

4.12.35 Relations: GATE CSE 2021 | Set 1 | Question: 43

A relation R is said to be circular if $a\text{R}b$ and $b\text{R}c$ together imply $c\text{R}a$.
Which of the following options is/are correct?

A. If a relation S is reflexive and symmetric, then S is an equivalence relation.
B. If a relation S is circular and symmetric, then S is an equivalence relation.
C. If a relation S is reflexive and circular, then S is an equivalence relation.
D. If a relation S is transitive and circular, then S is an equivalence relation.

gatecse-2021-set1 multiple-selects set-theory&algebra relations two-marks

Answer key

4.12.36 Relations: GATE CSE 2025 | Set 2 | Question: 32

Let $\mathcal{F}$ be the set of all functions from $\{1,\ldots,n\}$ to $\{0,1\}$. Define the binary relation $\leqslant$ on $\mathcal{F}$ as follows:
$\forall f,g \in \mathcal{F}, f \preccurlyeq g$ if and only if $\forall x \in \{1,\ldots,n\}, f(x) \leq g(x)$, where $0 \leq 1$.
Which of the following statement(s) is/are TRUE?

A. $\preccurlyeq$ is a symmetric relation
B. $(\mathcal{F}, \preccurlyeq)$ is a partial order
C. $(\mathcal{F}, \preccurlyeq)$ is a lattice
D. $\preccurlyeq$ is an equivalence relation

gatecse2025-set2 set-theory&algebra functions relations multiple-selects two-marks

Answer key

4.12.37 Relations: GATE CSE 2026 | Set 2 | Question: 16

Let R be a binary relation on the set $\{1,2,\ldots,10\}$, where $(x,y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R?

A. Reflexive
B. Symmetric
C. Transitive
D. Antisymmetric

gatecse-2026-set2 set-theory&algebra relations multiple-selects one-mark

Answer key

4.12.38 Relations: GATE IT 2004 | Question: 4

Let R_1 be a relation from $A=\{1,3,5,7\}$ to $B=\{2,4,6,8\}$ and R_2 be another relation from B to $C=\{1,2,3,4\}$ as defined below:

i. An element x in A is related to an element y in B (under R_1) if $x+y$ is divisible by 3.
ii. An element x in B is related to an element y in C (under R_2) if $x+y$ is even but not divisible by 3.

Which is the composite relation R_1R_2 from A to C?

A. $R_1R_2 = \{(1,2),(1,4),(3,3),(5,4),(7,3)\}$

B. $R_1R_2 = \{(1,2),(1,3),(3,2),(5,2),(7,3)\}$
C. $R_1R_2 = \{(1,2),(3,2),(3,4),(5,4),(7,2)\}$
D. $R_1R_2 = \{(3,2),(3,4),(5,1),(5,3),(7,1)\}$

gateit-2004 set-theory&algebra relations normal

Answer key☟

4.13 Set Theory (27)

4.13.1 Set Theory: GATE CSE 1993 | Question: 17

Out of a group of 21 persons, 9 eat vegetables, 10 eat fish and 7 eat eggs. 5 persons eat all three. How many persons eat at least two out of the three dishes?

gate1993 set-theory&algebra easy set-theory descriptive

Answer key☟

4.13.2 Set Theory: GATE CSE 1993 | Question: 8.3

Let S be an infinite set and $S_1 \ldots, S_n$ be sets such that $S_1 \cup S_2 \cup \cdots \cup S_n = S$. Then

A. at least one of the sets S_i is a finite set
B. not more than one of the sets S_i can be finite
C. at least one of the sets S_i is an infinite
D. not more than one of the sets S_i can be infinite
E. None of the above

gate1993 set-theory&algebra normal set-theory

Answer key☟

4.13.3 Set Theory: GATE CSE 1993 | Question: 8.4

Let A be a finite set of size n. The number of elements in the power set of $A \times A$ is:

A. 2^{2^n}
B. 2^{n^2}
C. $(2^n)^2$
D. $\left(2^2\right)^n$
E. None of the above

gate1993 set-theory&algebra easy set-theory

Answer key☟

4.13.4 Set Theory: GATE CSE 1994 | Question: 2.4

The number of subsets $\{1, 2, \ldots, n\}$ with odd cardinality is ___________

gate1994 set-theory&algebra easy set-theory fill-in-the-blanks

Answer key☟

4.13.5 Set Theory: GATE CSE 1995 | Question: 1.20

The number of elements in the power set $P(S)$ of the set $S = \{\{\emptyset\}, 1, \{2,3\}\}$ is:

A. 2
B. 4
C. 8
D. None of the above

gate1995 set-theory&algebra normal set-theory

Answer key☟

4.13.6 Set Theory: GATE CSE 1995 | Question: 25b

Determine the number of positive integers (≤ 720) which are not divisible by any of $2, 3$ or 5.

gate1995 set-theory&algebra set-theory numerical-answers

Answer key

4.13.7 Set Theory: GATE CSE 1996 | Question: 1.1

Let A and B be sets and let A^c and B^c denote the complements of the sets A and B. The set $(A-B) \cup (B-A) \cup (A \cap B)$ is equal to

A. $A \cup B$ B. $A^c \cup B^c$ C. $A \cap B$ D. $A^c \cap B^c$

gate1996 set-theory&algebra easy set-theory

Answer key

4.13.8 Set Theory: GATE CSE 1998 | Question: 2.4

In a room containing 28 people, there are 18 people who speak English, 15 people who speak Hindi and 22 people who speak Kannada. 9 persons speak both English and Hindi, 11 persons speak both Hindi and Kannada whereas 13 persons speak both Kannada and English. How many speak all three languages?

A. 9 B. 8 C. 7 D. 6

gate1998 set-theory&algebra easy set-theory

Answer key

4.13.9 Set Theory: GATE CSE 2000 | Question: 2.6

Let $P(S)$ denotes the power set of set S. Which of the following is always true?

A. $P(P(S)) = P(S)$
B. $P(S) \cap P(P(S)) = \{\emptyset\}$
C. $P(S) \cap S = P(S)$
D. $S \notin P(S)$

gatecse-2000 set-theory&algebra easy set-theory

Answer key

4.13.10 Set Theory: GATE CSE 2000 | Question: 6

Let S be a set of n elements $\{1, 2, \ldots, n\}$ and G a graph with 2^n vertices, each vertex corresponding to a distinct subset of S. Two vertices are adjacent iff the symmetric difference of the corresponding sets has exactly 2 elements. Note: The symmetric difference of two sets R_1 and R_2 is defined as $(R_1 \setminus R_2) \cup (R_2 \setminus R_1)$

Every vertex in G has the same degree. What is the degree of a vertex in G?

How many connected components does G have?

gatecse-2000 set-theory&algebra normal descriptive set-theory

Answer key

4.13.11 Set Theory: GATE CSE 2001 | Question: 2.2

Consider the following statements:

- S_1 : There exists infinite sets A, B, C such that $A \cap (B \cup C)$ is finite.
- S_2 : There exists two irrational numbers x and y such that $(x+y)$ is rational.

Which of the following is true about S_1 and S_2?

A. Only S_1 is correct
B. Only S_2 is correct
C. Both S_1 and S_2 are correct
D. None of S_1 and S_2 is correct

gatecse-2001 set-theory&algebra normal set-theory

Answer key

4.13.12 Set Theory: GATE CSE 2001 | Question: 3

a. Prove that powerset $(A \cap B) = \text{powerset}(A) \cap \text{powerset}(B)$

b. Let $\text{sum}(n) = 0 + 1 + 2 + \ldots\ldots + n$ for all natural numbers n. Give an induction proof to show that the following equation is true for all natural numbers m and n:

$$\text{sum}(m+n) = \text{sum}(m) + \text{sum}(n) + mn$$

gatecse-2001 set-theory&algebra normal set-theory descriptive

Answer key

4.13.13 Set Theory: GATE CSE 2005 | Question: 8

Let A, B and C be non-empty sets and let $X = (A - B) - C$ and $Y = (A - C) - (B - C)$. Which one of the following is TRUE?

A. $X = Y$
B. $X \subset Y$
C. $Y \subset X$
D. None of these

gatecse-2005 set-theory&algebra easy set-theory

Answer key

4.13.14 Set Theory: GATE CSE 2006 | Question: 22

Let E, F and G be finite sets. Let

- $X = (E \cap F) - (F \cap G)$ and
- $Y = (E - (E \cap G)) - (E - F)$.

Which one of the following is true?

A. $X \subset Y$
B. $X \supset Y$
C. $X = Y$
D. $X - Y \neq \emptyset$ and $Y - X \neq \emptyset$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.15 Set Theory: GATE CSE 2006 | Question: 24

Given a set of elements $N = 1, 2, \ldots, n$ and two arbitrary subsets $A \subseteq N$ and $B \subseteq N$, how many of the n! permutations π from N to N satisfy $\min(\pi(A)) = \min(\pi(B))$, where $\min(S)$ is the smallest integer in the set of integers S, and π(S) is the set of integers obtained by applying permutation π to each element of S?

A. $(n - |A \cup B|)|A||B|$
B. $(|A|^2 + |B|^2)n^2$
C. $n! \frac{|A \cap B|}{|A \cup B|}$
D. $\dfrac{|A \cap B|^2}{{}^{n}\mathrm{C}_{|A \cup B|}}$

gatecse-2006 set-theory&algebra normal set-theory

Answer key

4.13.16 Set Theory: GATE CSE 2008 | Question: 2

If P, Q, R are subsets of the universal set U, then

$$(P \cap Q \cap R) \cup (P^c \cap Q \cap R) \cup Q^c \cup R^c$$

is

A. $Q^c \cup R^c$
B. $P \cup Q^c \cup R^c$
C. $P^c \cup Q^c \cup R^c$
D. $\cup$

gatecse-2008 normal set-theory&algebra set-theory

Answer key

4.13.17 Set Theory: GATE CSE 2014 | Set 2 | Question: 50

Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U.

Consider the following two statements:

- $S1$: There is a subset of S that is larger than every other subset.
- $S2$: There is a subset of S that is smaller than every other subset.

Which one of the following is CORRECT?

A. Both $S1$ and $S2$ are true
B. $S1$ is true and $S2$ is false
C. $S2$ is true and $S1$ is false
D. Neither $S1$ nor $S2$ is true

gatecse-2014-set2 set-theory&algebra normal set-theory

Answer key☟

4.13.18 Set Theory: GATE CSE 2015 | Set 1 | Question: 16

For a set A, the power set of A is denoted by 2^A. If $A = \{5, \{6\}, \{7\}\}$, which of the following options are TRUE?

I. $\varnothing \in 2^A$
II. $\varnothing \subseteq 2^A$
III. $\{5, \{6\}\} \in 2^A$
IV. $\{5, \{6\}\} \subseteq 2^A$

A. I and III only
B. II and III only
C. I, II and III only
D. I, II and IV only

gatecse-2015-set1 set-theory&algebra set-theory normal

Answer key☟

4.13.19 Set Theory: GATE CSE 2015 | Set 2 | Question: 18

The cardinality of the power set of $\{0, 1, 2, \ldots, 10\}$ is _______

gatecse-2015-set2 set-theory&algebra set-theory easy numerical-answers

Answer key☟

4.13.20 Set Theory: GATE CSE 2015 | Set 3 | Question: 23

Suppose U is the power set of the set $S = \{1, 2, 3, 4, 5, 6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T. For any $T, R \in U$ let $T \backslash R$ be the set of all elements in T which are not in R. Which one of the following is true?

A. $\forall X \in U, (|X| = |X'|)$
B. $\exists X \in U, \exists Y \in U, (|X| = 5, |Y| = 5$ and $X \cap Y = \phi)$
C. $\forall X \in U, \forall Y \in U, (|X| = 2, |Y| = 3$ and $X \backslash Y = \phi)$
D. $\forall X \in U, \forall Y \in U, (X \backslash Y = Y' \backslash X')$

gatecse-2015-set3 set-theory&algebra set-theory normal

Answer key☟

4.13.21 Set Theory: GATE CSE 2016 | Set 2 | Question: 28

Consider a set U of 23 different compounds in a chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U. Consider the following statements:

I. Each compound in U \ S reacts with an odd number of compounds.
II. At least one compound in U \ S reacts with an odd number of compounds.
III. Each compound in U \ S reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

A. Only I
B. Only II
C. Only III
D. None.

gatecse-2016-set2 set-theory&algebra difficult set-theory

Answer key

4.13.22 Set Theory: GATE CSE 2017 | Set 1 | Question: 47

The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is ____________ .

gatecse-2017-set1 set-theory&algebra normal numerical-answers set-theory

Answer key

4.13.23 Set Theory: GATE CSE 2021 | Set 2 | Question: 37

For two n-dimensional real vectors P and Q, the operation $s(P, Q)$ is defined as follows:

$$s(P, Q) = \sum_{i=1}^{n} (P[i] \cdot Q[i])$$

Let $\mathcal{L}$ be a set of 10-dimensional non-zero real vectors such that for every pair of distinct vectors $P, Q \in \mathcal{L}$, $s(P, Q) = 0$. What is the maximum cardinality possible for the set $\mathcal{L}$?

A. 9
B. 10
C. 11
D. 100

gatecse-2021-set2 set-theory&algebra set-theory two-marks

Answer key

4.13.24 Set Theory: GATE IT 2004 | Question: 2

In a class of 200 students, 125 students have taken Programming Language course, 85 students have taken Data Structures course, 65 students have taken Computer Organization course; 50 students have taken both Programming Language and Data Structures, 35 students have taken both Programming Language and Computer Organization; 30 students have taken both Data Structures and Computer Organization, 15 students have taken all the three courses.

How many students have not taken any of the three courses?

A. 15
B. 20
C. 25
D. 30

gateit-2004 set-theory&algebra easy set-theory

Answer key

4.13.25 Set Theory: GATE IT 2005 | Question: 33

Let A be a set with n elements. Let C be a collection of distinct subsets of A such that for any two subsets S_1 and S_2 in C, either $S_1 \subset S_2$ or $S_2 \subset S_1$. What is the maximum cardinality of C?

A. n
B. $n + 1$
C. $2^{n-1} + 1$
D. $n!$

gateit-2005 set-theory&algebra normal set-theory

Answer key

4.13.26 Set Theory: GATE IT 2006 | Question: 23

Let P, Q and R be sets let Δ denote the symmetric difference operator defined as $P \Delta Q = (P \cup Q) - (P \cap Q)$. Using Venn diagrams, determine which of the following is/are TRUE?

I. $P \Delta (Q \cap R) = (P \Delta Q) \cap (P \Delta R)$
II. $P \cap (Q \cap R) = (P \cap Q) \Delta (P \Delta R)$

A. I only
B. II only
C. Neither I nor II
D. Both I and II

gateit-2006 set-theory&algebra normal set-theory

Answer key

4.13.27 Set Theory: GATE IT 2006 | Question: 24

What is the cardinality of the set of integers X defined below?

$X = \{n \mid 1 \leq n \leq 123, n$ is not divisible by either $2, 3$ or $5\}$

A. 28 B. 33 C. 37 D. 44

gateit-2006 set-theory&algebra normal set-theory

Answer key

Answer Keys

4.1.1	N/A	4.1.2	N/A	4.1.3	C	4.1.4	D	4.1.5	A
4.1.6	A	4.1.7	B	4.1.8	A	4.2.1	True	4.2.2	D
4.3.1	N/A	4.3.2	N/A	4.3.3	N/A	4.3.4	B	4.3.5	A
4.3.6	C	4.3.7	N/A	4.3.8	C	4.3.9	A	4.3.10	N/A
4.3.11	A	4.3.12	B	4.3.13	B	4.3.14	D	4.3.15	B
4.3.16	C	4.3.17	C	4.3.18	16	4.3.19	D	4.3.20	B
4.3.21	A	4.3.22	36	4.3.23	0.95	4.3.24	2	4.3.25	B;C
4.3.26	B;C;D	4.3.27	2	4.3.28	10:10	4.3.29	C	4.3.30	D
4.4.1	N/A	4.4.2	N/A	4.4.3	N/A	4.4.4	N/A	4.4.5	C
4.4.6	D	4.4.7	N/A	4.4.8	C	4.4.9	C	4.4.10	D
4.4.11	N/A	4.4.12	N/A	4.4.13	N/A	4.4.14	B	4.4.15	A
4.4.16	D	4.4.17	C	4.4.18	A	4.4.19	C	4.4.20	A
4.4.21	A	4.4.22	C	4.4.23	A	4.4.24	5	4.4.25	4
4.4.26	42	4.4.27	B	4.4.28	7	4.4.29	B	4.4.30	A;B;C
4.4.31	A;D	4.4.32	B;D	4.4.33	4	4.5.1	B	4.6.1	N/A
4.6.2	N/A	4.6.3	D	4.6.4	5	4.6.5	B	4.6.6	N/A
4.6.7	B	4.6.8	D	4.6.9	0	4.6.10	A	4.7.1	N/A
4.7.2	N/A	4.8.1	24	4.8.2	8	4.8.3	36	4.8.4	D
4.8.5	D	4.8.6	C	4.8.7	N/A	4.9.1	D	4.10.1	N/A
4.10.2	E	4.10.3	B	4.10.4	A	4.10.5	N/A	4.10.6	D
4.10.7	A	4.10.8	C	4.10.9	5	4.10.10	D	4.11.1	B
4.11.2	D	4.11.3	A	4.11.4	8	4.12.1	False	4.12.2	N/A
4.12.3	N/A	4.12.4	N/A	4.12.5	N/A	4.12.6	N/A	4.12.7	D
4.12.8	B	4.12.9	N/A	4.12.10	N/A	4.12.11	A	4.12.12	B
4.12.13	C	4.12.14	N/A	4.12.15	N/A	4.12.16	B	4.12.17	C
4.12.18	A;C	4.12.19	N/A	4.12.20	C	4.12.21	B	4.12.22	D
4.12.23	N/A	4.12.24	B	4.12.25	C	4.12.26	D	4.12.27	A
4.12.28	B	4.12.29	D	4.12.30	C	4.12.31	D	4.12.32	C
4.12.33	B	4.12.34	0.1249:0.1259	4.12.35	C	4.12.36	B;C	4.12.37	A;B;C
4.12.38	C	4.13.1	N/A	4.13.2	C	4.13.3	B	4.13.4	N/A
4.13.5	C	4.13.6	192	4.13.7	A	4.13.8	D	4.13.9	B

4.13.10	N/A
4.13.15	C
4.13.20	D
4.13.25	B

4.13.11	C
4.13.16	D
4.13.21	B
4.13.26	C

4.13.12	N/A
4.13.17	A
4.13.22	271
4.13.27	B

4.13.13	A
4.13.18	C
4.13.23	B

4.13.14	C
4.13.19	2048
4.13.24	C

5 Engineering Mathematics: Calculus (69)

Syllabus: Limits, Continuity, and Differentiability, Maxima and minima, Mean value theorem, Integration.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	1	1	1	1	1	1	2	1	1	1	1
2 Marks Count	1	1	0	0	0	0	0	0	0	0	0
Total Marks	3	3	1	1	1	1	2	1	1	1	**1**

Welcome to the "Engineering Mathematics: Calculus" chapter, a cornerstone of your GATE Computer Science preparation. This section is designed as a comprehensive, exam-focused reference to equip you with the essential concepts, formulas, and problem-solving strategies required to excel in the calculus portion of the GATE exam. Calculus is not merely a collection of mathematical techniques; it is the language of change and optimization, fundamental to understanding algorithms, data structures, probability, and various computational models. A solid grasp of calculus concepts will not only help you score well in the dedicated Engineering Mathematics section but also enhance your analytical abilities for other subjects. Typically, Engineering Mathematics accounts for 8-10% of the total GATE marks, with Calculus being a significant component of this weightage. Questions usually range from direct application of formulas and theorems to conceptual problems involving limits, continuity, differentiation, and integration, often requiring a blend of analytical reasoning and careful calculation. Expect a mix of Multiple Choice Questions (MCQ), Multiple Select Questions (MSQ), and Numerical Answer Type (NAT) questions.

Topic-wise Key Concepts

Limits

Definition and Core Idea: A limit describes the behavior of a function as its input approaches a certain value. It tells us what value the function 'tends to' or 'approaches' as the independent variable gets arbitrarily close to a specific point, without necessarily reaching it.

Important Formulas, Theorems, and Results:

1. **Algebra of Limits:** If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} g(x) = M$, then:
    - $\lim_{x\to a}[f(x) \pm g(x)] = L \pm M$
    - $\lim_{x\to a}[f(x) \cdot g(x)] = L \cdot M$
    - $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$, provided $M \neq 0$
    - $\lim_{x\to a}[c \cdot f(x)] = c \cdot L$
2. **Standard Limits:**
    - $\lim_{x\to 0} \frac{\sin x}{x} = 1$
    - $\lim_{x\to 0} \frac{\tan x}{x} = 1$
    - $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
    - $\lim_{x\to 0} \frac{e^x-1}{x} = 1$
    - $\lim_{x\to 0} \frac{a^x-1}{x} = \ln a$
    - $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
    - $\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$
    - $\lim_{x\to\infty}(1+\frac{1}{x})^x = e$
    - $\lim_{x\to 0}(1+x)^{1/x} = e$
    - $\lim_{x\to\infty}(1+\frac{k}{x})^x = e^k$
3. **L'Hopital's Rule:** If $\lim_{x\to a} \frac{f(x)}{g(x)}$ is of the indeterminate form $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$, provided the latter limit exists. This can be applied repeatedly.
4. **Squeeze Theorem (Sandwich Theorem):** If $g(x) \leq f(x) \leq h(x)$ for all x in an open interval containing c (except possibly at c itself), and if $\lim_{x\to c} g(x) = L$ and $\lim_{x\to c} h(x) = L$, then $\lim_{x\to c} f(x) = L$.

Key Properties and Identities:

- Indeterminate forms: $\frac{0}{0}, \frac{\infty}{\infty}, 0\cdot\infty, \infty-\infty, 1^\infty, 0^0, \infty^0$. These require special techniques like L'Hopital's Rule or algebraic manipulation.

- One-sided limits: For a limit to exist, the left-hand limit ($\lim_{x \to a^-} f(x)$) and the right-hand limit ($\lim_{x \to a^+} f(x)$) must be equal.

Common Pitfalls or Tricky Points:

- Applying L'Hopital's Rule when the limit is not an indeterminate form.
- Incorrectly handling one-sided limits, especially for piecewise functions or functions with square roots.
- Errors in algebraic simplification before applying standard limits.
- Misinterpreting limits involving infinity (e.g., $\frac{1}{\infty} = 0$, but $\infty \cdot 0$ is indeterminate).

Standard Problem-Solving Techniques or Shortcuts:

- **Direct Substitution:** Always try this first. If it yields a finite value, that's the limit.
- **Factorization and Cancellation:** For $\frac{0}{0}$ forms involving polynomials, factorize the numerator and denominator to cancel common terms.
- **Rationalization:** For expressions involving square roots, multiply by the conjugate to simplify.
- **Using Standard Limits:** Manipulate the expression to fit one of the standard limit forms.
- **L'Hopital's Rule:** Apply for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ forms. Remember to differentiate numerator and denominator separately.
- **Transforming Indeterminate Forms:**
  - $0 \cdot \infty \implies \frac{0}{1/\infty}$ or $\frac{\infty}{1/0}$
  - $\infty - \infty \implies$ combine terms, rationalize, or find a common denominator.
  - $1^\infty, 0^0, \infty^0 \implies$ take natural logarithm: If $y = [f(x)]^{g(x)}$, then $\ln y = g(x) \ln f(x)$. Then find $\lim \ln y$ and exponentiate the result.

Continuity

Definition and Core Idea: A function is continuous at a point if its graph has no breaks, jumps, or holes at that point. Informally, you can draw the graph without lifting your pen. Formally, it means the function's value at the point is equal to its limit as the input approaches that point.

Important Formulas, Theorems, and Results:

1. **Conditions for Continuity at a Point $x = a$:** A function $f(x)$ is continuous at $x = a$ if and only if all three conditions are met:
   a. $f(a)$ is defined (the function exists at a).
   b. $\lim_{x \to a} f(x)$ exists (the limit exists at a). This implies $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$.
   c. $\lim_{x \to a} f(x) = f(a)$ (the limit equals the function value).
2. **Continuity on an Interval:** A function is continuous on an open interval (a, b) if it is continuous at every point in the interval. It is continuous on a closed interval $[a, b]$ if it is continuous on (a, b), continuous from the right at a ($\lim_{x \to a^+} f(x) = f(a)$), and continuous from the left at b ($\lim_{x \to b^-} f(x) = f(b)$).
3. **Intermediate Value Theorem (IVT):** If f is continuous on the closed interval $[a, b]$ and k is any number between $f(a)$ and $f(b)$, then there exists at least one number c in (a, b) such that $f(c) = k$.
4. **Extreme Value Theorem (EVT):** If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value and an absolute minimum value on that interval.

Key Properties and Identities:

- The sum, difference, product, and quotient (where the denominator is non-zero) of continuous functions are continuous.
- The composition of continuous functions is continuous: If g is continuous at a and f is continuous at $g(a)$, then $(f \circ g)(x) = f(g(x))$ is continuous at a.
- Polynomials, exponential functions (e^x, a^x), logarithmic functions ($\ln x, \log_a x$), trigonometric functions ($\sin x, \cos x$), and rational functions (where the denominator is non-zero) are continuous in their domains.

Common Pitfalls or Tricky Points:

- **Piecewise Functions:** Pay close attention to the points where the definition of the function changes. These are the critical points to check for continuity.
- **Removable Discontinuities:** Occur when $\lim_{x \to a} f(x)$ exists but is not equal to $f(a)$, or $f(a)$ is undefined (e.g., holes in the graph).
- **Jump Discontinuities:** Occur when the left-hand and right-hand limits exist but are not equal (e.g., step functions).

- **Infinite Discontinuities:** Occur when one or both of the one-sided limits are infinite (e.g., vertical asymptotes).
- Forgetting to check all three conditions for continuity.

Standard Problem-Solving Techniques or Shortcuts:

- For piecewise functions, evaluate the function value at the transition point, and calculate both the left-hand and right-hand limits at that point. Equate them to find unknown constants.
- Identify the domain of the function first. Discontinuities often occur at points where the function is undefined (e.g., denominator is zero, negative under square root).
- For functions involving absolute values, rewrite them as piecewise functions.

Differentiation

Definition and Core Idea: Differentiation is the process of finding the derivative of a function. The derivative represents the instantaneous rate of change of a function with respect to its independent variable. Geometrically, it gives the slope of the tangent line to the function's graph at a given point.

Important Formulas, Theorems, and Results:

1. **Definition of Derivative:** $f'(x) = \lim_{h \to 0} \frac{f(x+h)-f(x)}{h}$
2. **Basic Differentiation Rules:**
   - $\frac{d}{dx}(c) = 0$ (where c is a constant)
   - $\frac{d}{dx}(x^n) = nx^{n-1}$
   - $\frac{d}{dx}(e^x) = e^x$
   - $\frac{d}{dx}(a^x) = a^x \ln a$
   - $\frac{d}{dx}(\ln x) = \frac{1}{x}$
   - $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$
   - $\frac{d}{dx}(\sin x) = \cos x$
   - $\frac{d}{dx}(\cos x) = -\sin x$
   - $\frac{d}{dx}(\tan x) = \sec^2 x$
   - $\frac{d}{dx}(\cot x) = -\csc^2 x$
   - $\frac{d}{dx}(\sec x) = \sec x \tan x$
   - $\frac{d}{dx}(\csc x) = -\csc x \cot x$
   - $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
   - $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
   - $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
3. **Rules of Differentiation:**
   - **Constant Multiple Rule:** $\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$
   - **Sum/Difference Rule:** $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$
   - **Product Rule:** $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$
   - **Quotient Rule:** $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$
   - **Chain Rule:** $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$
4. **Implicit Differentiation:** Used when y is not explicitly defined as a function of x. Differentiate both sides of the equation with respect to x, treating y as a function of x (i.e., use the chain rule for terms involving y).
5. **Rolle's Theorem:** If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then there exists at least one $c \in (a, b)$ such that $f'(c) = 0$.
6. **Mean Value Theorem (Lagrange's MVT):** If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists at least one $c \in (a, b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.
7. **Cauchy's Mean Value Theorem:** If f and g are continuous on $[a, b]$ and differentiable on (a, b), and $g'(x) \neq 0$ for any $x \in (a, b)$, then there exists at least one $c \in (a, b)$ such that $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$.

Key Properties and Identities:

- Differentiability implies continuity, but continuity does not imply differentiability (e.g., $|x|$ at $x = 0$).

- Higher-order derivatives: $f''(x) = \frac{d}{dx}(f'(x))$, $f'''(x) = \frac{d}{dx}(f''(x))$, etc.

Common Pitfalls or Tricky Points:

- Forgetting the chain rule, especially with composite functions like $\sin(2x)$ or e^{x^2}.
- Errors in applying the product or quotient rule.
- Mistakes in implicit differentiation, particularly when differentiating terms involving y (e.g., $\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}$).
- Confusing the derivative of a^x with x^a.
- Not simplifying expressions before differentiating when possible.

Standard Problem-Solving Techniques or Shortcuts:

- **Logarithmic Differentiation:** For functions of the form $y = [f(x)]^{g(x)}$ or complex products/quotients, take the natural logarithm of both sides before differentiating implicitly.
- **Parametric Differentiation:** If $x = f(t)$ and $y = g(t)$, then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.
- **Derivative of Inverse Function:** If $y = f^{-1}(x)$, then $\frac{dy}{dx} = \frac{1}{f'(y)}$.
- Practice recognizing common derivative patterns for speed.

Maxima Minima

Definition and Core Idea: Maxima and minima (collectively called extrema) refer to the highest and lowest values a function attains within a given interval or its entire domain. A local maximum/minimum is the highest/lowest point in its immediate neighborhood, while a global (absolute) maximum/minimum is the highest/lowest point over the entire domain/interval.

Important Formulas, Theorems, and Results:

1. **Critical Points:** A critical point of a function $f(x)$ is a point c in the domain of f where either $f'(c) = 0$ or $f'(c)$ is undefined. Local extrema can only occur at critical points or endpoints of a closed interval.
2. **First Derivative Test:** Let c be a critical point of a continuous function f.
   - If $f'(x)$ changes from positive to negative at c, then f has a local maximum at c.
   - If $f'(x)$ changes from negative to positive at c, then f has a local minimum at c.
   - If $f'(x)$ does not change sign at c, then f has neither a local maximum nor a local minimum at c.
3. **Second Derivative Test:** Let c be a critical point where $f'(c) = 0$ and $f''(c)$ exists.
   - If $f''(c) > 0$, then f has a local minimum at c.
   - If $f''(c) < 0$, then f has a local maximum at c.
   - If $f''(c) = 0$, the test is inconclusive (use the First Derivative Test).
4. **Concavity:**
   - If $f''(x) > 0$ on an interval, f is concave up (graph opens upwards).
   - If $f''(x) < 0$ on an interval, f is concave down (graph opens downwards).
5. **Inflection Point:** A point where the concavity of the function changes. This occurs where $f''(x) = 0$ or $f''(x)$ is undefined, and $f''(x)$ changes sign.

Key Properties and Identities:

- For global extrema on a closed interval $[a, b]$, evaluate $f(x)$ at all critical points in (a, b) and at the endpoints a and b. The largest of these values is the absolute maximum, and the smallest is the absolute minimum.

Common Pitfalls or Tricky Points:

- Forgetting to check endpoints when finding global extrema on a closed interval.
- Misinterpreting the results of the second derivative test, especially when $f''(c) = 0$.
- Not considering points where $f'(x)$ is undefined as critical points.
- Errors in algebraic manipulation while finding critical points or second derivatives.
- Confusing local extrema with global extrema.

Standard Problem-Solving Techniques or Shortcuts:

- **Optimization Problems:**
  1. Understand the problem and identify the quantity to be maximized or minimized.

2. Draw a diagram if helpful.
3. Introduce variables and write the primary equation for the quantity to be optimized.
4. Write a secondary equation (constraint) relating the variables.
5. Use the secondary equation to express the primary equation in terms of a single independent variable.
6. Find the domain of this function.
7. Differentiate the function, find critical points, and apply the First or Second Derivative Test.
8. Verify the solution makes sense in the context of the problem.

- For simple functions, sometimes the graph can quickly indicate extrema.

Integration

Definition and Core Idea: Integration is the reverse process of differentiation, also known as antidifferentiation. It finds the function whose derivative is the given function. Geometrically, definite integration calculates the area under the curve of a function between two specified limits.

Important Formulas, Theorems, and Results:

1. **Basic Integration Formulas (Indefinite Integrals):**
   - $\int k\,dx = kx + C$
   - $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$, for $n \neq -1$
   - $\int \frac{1}{x}\,dx = \ln|x| + C$
   - $\int e^x\,dx = e^x + C$
   - $\int a^x\,dx = \frac{a^x}{\ln a} + C$
   - $\int \sin x\,dx = -\cos x + C$
   - $\int \cos x\,dx = \sin x + C$
   - $\int \sec^2 x\,dx = \tan x + C$
   - $\int \csc^2 x\,dx = -\cot x + C$
   - $\int \sec x \tan x\,dx = \sec x + C$
   - $\int \csc x \cot x\,dx = -\csc x + C$
   - $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
   - $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
   - $\int \frac{1}{x\sqrt{x^2-a^2}}\,dx = \frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right) + C$
2. **Properties of Indefinite Integrals:**
   - $\int [f(x) \pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$
   - $\int cf(x)\,dx = c\int f(x)\,dx$
3. **Integration by Substitution (u-substitution):** If $u = g(x)$, then $du = g'(x)\,dx$. $\int f(g(x))g'(x)\,dx = \int f(u)\,du$.
4. **Integration by Parts:** $\int u\,dv = uv - \int v\,du$. Often remembered by the mnemonic "LIATE" (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential) for choosing u.

Key Properties and Identities:

- Always include the constant of integration $+C$ for indefinite integrals.
- Trigonometric identities are often crucial for simplifying integrands before integration.

Common Pitfalls or Tricky Points:

- Forgetting the constant of integration $+C$.
- Incorrectly choosing u and dv in integration by parts.
- Errors in algebraic manipulation or trigonometric identities.
- Not recognizing when substitution is appropriate.
- Confusing integration formulas with differentiation formulas (e.g., $\int \frac{1}{x}dx$ vs. $\frac{d}{dx}\ln x$).

Standard Problem-Solving Techniques or Shortcuts:

- **Direct Integration:** If the integrand matches a basic formula.
- **Substitution:** Look for a function and its derivative within the integrand. Let the inner function be$u$.
- **Integration by Parts:** Use for products of functions (e.g., $xe^x, x\sin x, \ln x$).
- **Partial Fractions:** For rational functions (polynomials divided by polynomials), decompose into simpler fractions that are easier to integrate. (Briefly mention as it's a common technique).

- **Trigonometric Identities:** Use identities to simplify powers of trig functions or products of trig functions.

Definite Integral

Definition and Core Idea: A definite integral represents the net signed area between the graph of a function and the x-axis over a specified interval $[a, b]$. It is evaluated by finding the antiderivative of the function and then evaluating it at the upper and lower limits of integration.

Important Formulas, Theorems, and Results:

1. **Fundamental Theorem of Calculus (Part 1):** If f is continuous on $[a, b]$, then the function $G(x) = \int_a^x f(t)\,dt$ has a derivative at every point in (a, b), and $G'(x) = f(x)$.
2. **Fundamental Theorem of Calculus (Part 2):** If f is continuous on $[a, b]$ and F is any antiderivative of f on $[a, b]$, then $\int_a^b f(x)\,dx = F(b) - F(a)$.
3. **Properties of Definite Integrals:**
    - $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
    - $\int_a^a f(x)\,dx = 0$
    - $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$
    - $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
    - $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $a < c < b$
    - $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (useful for simplifying integrals)
    - **Even/Odd Functions:**
        - If $f(x)$ is an even function ($f(-x) = f(x)$), then $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$.
        - If $f(x)$ is an odd function ($f(-x) = -f(x)$), then $\int_{-a}^a f(x)\,dx = 0$.

Key Properties and Identities:

- The definite integral evaluates to a numerical value, not a function.
- The constant of integration $+C$ is not included in definite integrals because it cancels out ($F(b) + C - (F(a) + C) = F(b) - F(a)$).

Common Pitfalls or Tricky Points:

- Incorrectly applying the limits of integration (upper limit minus lower limit).
- Sign errors when evaluating $F(b) - F(a)$.
- Forgetting to change the limits of integration when using substitution for definite integrals.
- Not recognizing even/odd function properties to simplify integrals over symmetric intervals.
- Improper integrals (involving infinite limits or discontinuities within the interval) are less common in GATE CS but good to be aware of.

Standard Problem-Solving Techniques or Shortcuts:

- **Direct Application of FTC:** Find the antiderivative and evaluate at the limits.
- **Substitution:** If using substitution, either change the limits of integration to be in terms of u or revert back to x before applying the original limits.
- **Properties of Definite Integrals:** Use properties like linearity, interval splitting, and even/odd function rules to simplify the integral before evaluation.
- **Symmetry:** For integrals over $[-a, a]$, check if the integrand is even or odd.

Polynomials

Definition and Core Idea: A polynomial is an expression consisting of variables and coefficients, involving only the operations of addition, subtraction, multiplication, and non-negative integer exponents of variables. They are fundamental building blocks in calculus as they are continuous and differentiable everywhere.

Important Formulas, Theorems, and Results:

1. **General Form:** $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, where a_i are coefficients and n is a non-negative integer (degree).
2. **Remainder Theorem:** If a polynomial $P(x)$ is divided by $(x - a)$, the remainder is $P(a)$.

3. **Factor Theorem:** $(x-a)$ is a factor of a polynomial $P(x)$ if and only if $P(a)=0$ (i.e., a is a root of the polynomial).
4. **Fundamental Theorem of Algebra:** Every non-constant single-variable polynomial with complex coefficients has at least one complex root. Consequently, a polynomial of degree n has exactly n roots (counting multiplicity) in the complex numbers.
5. **Vieta's Formulas (for quadratic and cubic):**
   - For $ax^2+bx+c=0$ with roots α, β:
     - $\alpha+\beta=-\frac{b}{a}$
     - $\alpha\beta=\frac{c}{a}$
   - For $ax^3+bx^2+cx+d=0$ with roots α, β, γ:
     - $\alpha+\beta+\gamma=-\frac{b}{a}$
     - $\alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a}$
     - $\alpha\beta\gamma=-\frac{d}{a}$

Key Properties and Identities:

- Polynomials are continuous and infinitely differentiable everywhere.
- The degree of a polynomial determines the maximum number of roots it can have.
- If a polynomial has real coefficients, then any complex roots must occur in conjugate pairs.

Common Pitfalls or Tricky Points:

- Errors in polynomial division or synthetic division.
- Misapplying Vieta's formulas (e.g., sign errors).
- Not considering multiplicity of roots.
- Confusing roots with factors.

Standard Problem-Solving Techniques or Shortcuts:

- **Synthetic Division:** A quick method for dividing a polynomial by a linear factor $(x-a)$.
- **Rational Root Theorem:** Helps find potential rational roots of a polynomial with integer coefficients.
- **Factoring:** Look for common factors, use grouping, or quadratic formula for quadratic factors.
- **Derivative for Repeated Roots:** If a is a root of $P(x)$ with multiplicity $k>1$, then a is also a root of $P'(x)$ with multiplicity $k-1$.

Summation

Definition and Core Idea: Summation is the process of adding a sequence of numbers or terms. It is often represented using sigma notation ($\sum$). In calculus, summation is foundational to understanding series, Riemann sums (for integration), and numerical methods.

Important Formulas, Theorems, and Results:

1. **Arithmetic Progression (AP) Sum:** For an AP with first term a, common difference d, and n terms:
   - $S_n=\frac{n}{2}[2a+(n-1)d]$
   - $S_n=\frac{n}{2}[a+l]$, where l is the last term.
2. **Geometric Progression (GP) Sum:** For a GP with first term a, common ratio r, and n terms:
   - $S_n=\frac{a(r^n-1)}{r-1}$, for $r\neq 1$
   - $S_n=\frac{a(1-r^n)}{1-r}$, for $r\neq 1$
   - **Sum to Infinity for GP:** $S_\infty=\frac{a}{1-r}$, for $|r|<1$
3. **Standard Summation Formulas:**
   - $\sum_{i=1}^{n} c=nc$
   - $\sum_{i=1}^{n} i=\frac{n(n+1)}{2}$
   - $\sum_{i=1}^{n} i^2=\frac{n(n+1)(2n+1)}{6}$
   - $\sum_{i=1}^{n} i^3=\left(\frac{n(n+1)}{2}\right)^2$

Key Properties and Identities:

- **Linearity of Summation:**

    - $\sum_{i=1}^{n}(a_i \pm b_i) = \sum_{i=1}^{n} a_i \pm \sum_{i=1}^{n} b_i$
    - $\sum_{i=1}^{n} c \cdot a_i = c \cdot \sum_{i=1}^{n} a_i$
- **Telescoping Sums:** Sums where intermediate terms cancel out, e.g., $\sum_{i=1}^{n}(a_i - a_{i+1})$.

Common Pitfalls or Tricky Points:

- Off-by-one errors in summation limits.
- Confusing AP and GP formulas.
- Incorrectly applying the sum to infinity formula (only valid for $|r| < 1$).
- Errors in algebraic manipulation when simplifying complex summations.

Standard Problem-Solving Techniques or Shortcuts:

- **Identify Series Type:** Determine if it's an AP, GP, or a combination.
- **Use Standard Formulas:** Apply the formulas for $\sum i, \sum i^2, \sum i^3$.
- **Split and Combine:** Use linearity to break down complex summations into simpler ones.
- **Method of Differences (Telescoping Sums):** Express each term as a difference of two consecutive terms to simplify the sum.
- **Generating Functions:** (Advanced, less common for direct GATE CS questions, but good for understanding series).

Quick Formula Reference

Limits

- $\lim_{x\to 0} \frac{\sin x}{x} = 1$
- $\lim_{x\to 0} \frac{\tan x}{x} = 1$
- $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0} \frac{e^x-1}{x} = 1$
- $\lim_{x\to 0} \frac{a^x-1}{x} = \ln a$
- $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to a} \frac{x^n-a^n}{x-a} = na^{n-1}$
- $\lim_{x\to \infty}(1+\frac{1}{x})^x = e$
- $\lim_{x\to 0}(1+x)^{1/x} = e$
- L'Hopital's Rule: If $\frac{f(x)}{g(x)}$ is $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then $\lim \frac{f(x)}{g(x)} = \lim \frac{f'(x)}{g'(x)}$.

Continuity

- Continuous at $x = a$ if: $f(a)$ defined, $\lim_{x\to a} f(x)$ exists, and $\lim_{x\to a} f(x) = f(a)$.

Differentiation

Let u, v be functions of x, c be a constant.

- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(a^x) = a^x \ln a$
- $\frac{d}{dx}(\ln x) = \frac{1}{x}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(cu) = c\frac{du}{dx}$
- $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$
- $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

- $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
- $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain Rule)
- Rolle's Theorem: $f(a) = f(b) \implies \exists c \in (a, b) \text{ s.t. } f'(c) = 0$
- Mean Value Theorem: $\exists c \in (a, b) \text{ s.t. } f'(c) = \frac{f(b)-f(a)}{b-a}$

Maxima Minima

- Critical points: $f'(x) = 0$ or $f'(x)$ undefined.
- First Derivative Test: Sign change of $f'(x)$ at critical point.
- Second Derivative Test: If $f'(c) = 0$: $f''(c) > 0 \implies$ local min; $f''(c) < 0 \implies$ local max.

Integration (Indefinite)

- $\int k\,dx = kx + C$
- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$, for $n \neq -1$
- $\int \frac{1}{x}\,dx = \ln|x| + C$
- $\int e^x\,dx = e^x + C$
- $\int a^x\,dx = \frac{a^x}{\ln a} + C$
- $\int \sin x\,dx = -\cos x + C$
- $\int \cos x\,dx = \sin x + C$
- $\int \sec^2 x\,dx = \tan x + C$
- $\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\left(\frac{x}{a}\right) + C$
- $\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$
- Integration by Parts: $\int u\,dv = uv - \int v\,du$

Definite Integral

- Fundamental Theorem of Calculus: $\int_a^b f(x)\,dx = F(b) - F(a)$ where $F'(x) = f(x)$.
- $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$
- $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$
- $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$
- Even function ($f(-x) = f(x)$): $\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx$
- Odd function ($f(-x) = -f(x)$): $\int_{-a}^a f(x)\,dx = 0$

Polynomials

- Remainder Theorem: $P(x)$ divided by $(x-a)$ has remainder $P(a)$.
- Factor Theorem: $(x-a)$ is a factor of $P(x)$ iff $P(a) = 0$.
- Vieta's Formulas (for $ax^2 + bx + c = 0$): $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.

Summation

- AP Sum: $S_n = \frac{n}{2}[2a + (n-1)d]$ or $S_n = \frac{n}{2}[a + l]$
- GP Sum: $S_n = \frac{a(r^n-1)}{r-1}$
- GP Sum to Infinity: $S_\infty = \frac{a}{1-r}$ for $|r| < 1$
- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2}\right)^2$

Important Tips for GATE

1. **Master the Fundamentals:** Calculus builds upon basic algebra, trigonometry, and functions. Ensure your foundation in these areas is strong to avoid silly mistakes.
2. **Practice with PYQs:** Solve previous year's GATE questions extensively. This helps you understand the typical question patterns, difficulty levels, and time constraints.
3. **Understand Concepts, Don't Just Memorize:** While formulas are crucial, knowing *why* they work (e.g., the geometric interpretation of derivatives and integrals) will help you tackle trickier conceptual questions and apply them correctly in unfamiliar scenarios.
4. **Pay Attention to Domains and Conditions:** For limits, continuity, and differentiability, always check the domain of the function and any specific conditions (e.g., $g(x) \neq 0$ for quotient rule, $|r| < 1$ for infinite GP sum). Piecewise functions are common traps.
5. **Be Careful with Signs and Constants:** A common source of error in differentiation and integration is incorrect signs or forgetting the constant of integration ($+C$) for indefinite integrals. Double-check your calculations.
6. **Time Management:** Some calculus problems can be lengthy. Learn to identify problems that might take too much time and mark them for review. Practice solving problems within a strict time limit. Shortcuts like L'Hopital's rule or properties of definite integrals can save valuable time.
7. **Visualize Functions:** For problems involving maxima/minima or definite integrals, sketching a rough graph of the function can often provide intuition and help verify your analytical results.
8. **Review Properties of Even/Odd Functions:** These properties are frequently tested in definite integral problems, especially over symmetric intervals like $[-a, a]$, and can significantly simplify calculations.

5.1 Continuity (11)

✍ **Practice Test:** Test 1 (14Q)

5.1.1 Continuity: GATE CSE 1996 | Question: 3

Let f be a function defined by

$$f(x)=\begin{cases} x^2 & \text{for } x \leq 1 \\ ax^2+bx+c & \text{for } 1 \quad ltx \leq 2 \\ x+d & \text{for } x > 2 \end{cases}$$

Find the values for the constants a, b, c and d so that f is continuous and differentiable everywhere on the real line.

gate1996 calculus continuity differentiation normal descriptive

Answer key☟

5.1.2 Continuity: GATE CSE 1998 | Question: 1.4

Consider the function $y = |x|$ in the interval $[-1, 1]$. In this interval, the function is

A. continuous and differentiable
B. continuous but not differentiable
C. differentiable but not continuous
D. neither continuous nor differentiable

gate1998 calculus continuity differentiation easy

Answer key☟

5.1.3 Continuity: GATE CSE 2007 | Question: 1

Consider the following two statements about the function $f(x) = |x|$:

- P. $f(x)$ is continuous for all real values of x.
- Q. $f(x)$ is differentiable for all real values of x .

Which of the following is **TRUE**?

A. P is true and Q is false.
B. P is false and Q is true.
C. Both P and Q are true.
D. Both P and Q are false.

gatecse-2007 calculus continuity differentiation easy

Answer key

5.1.4 Continuity: GATE CSE 2013 | Question: 22

Which one of the following functions is continuous at $x = 3$?

A. $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x - 1, & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$
B. $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8 - x, & \text{if } x \neq 3 \end{cases}$
C. $f(x) = \begin{cases} x + 3, & \text{if } x \leq 3 \\ x - 4, & \text{if } x > 3 \end{cases}$
D. $f(x) = \begin{cases} \frac{1}{x^3 - 27}, & \text{if } x \neq 3 \end{cases}$

gatecse-2013 calculus continuity normal

Answer key

5.1.5 Continuity: GATE CSE 2014 | Set 1 | Question: 47

A function $f(x)$ is continuous in the interval $[0, 2]$. It is known that $f(0) = f(2) = -1$ and $f(1) = 1$. Which one of the following statements must be true?

A. There exists a y in the interval $(0, 1)$ such that $f(y) = f(y + 1)$
B. For every y in the interval $(0, 1)$,$f(y)$ = $f(2 - y)$
C. The maximum value of the function in the interval $(0, 2)$ is 1
D. There exists a y in the interval $(0, 1)$ such that $f(y)$ = $-f(2 - y)$

gatecse-2014-set1 calculus continuity tricky

Answer key

5.1.6 Continuity: GATE CSE 2015 | Set 2 | Question: 26

Let $f(x) = x^{-\left(\frac{1}{3}\right)}$ and A denote the area of region bounded by $f(x)$ and the X-axis, when x varies from -1 to 1. Which of the following statements is/are TRUE?

I. f is continuous in $[-1, 1]$
II. f is not bounded in $[-1, 1]$
III. A is nonzero and finite

A. II only
B. III only
C. II and III only
D. I, II and III

gatecse-2015-set2 calculus continuity functions normal

Answer key

5.1.7 Continuity: GATE CSE 2021 | Set 2 | Question: 25

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous function on the interval $[-3, 3]$ and a differentiable function in the interval $(-3, 3)$ such that for every x in the interval, $f'(x) \leq 2$. If $f(-3) = 7$, then $f(3)$ is at most ________

gatecse-2021-set2 numerical-answers calculus continuity one-mark

Answer key

5.1.8 Continuity: GATE CSE 2026 | Set 1 | Question: 22

Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined as follows:

$$f(x) = \begin{cases} c_1 e^x - c_2 \log_e\left(\frac{1}{x}\right), & \text{if } x > 0 \\ 3 & \text{otherwise} \end{cases}$$

where $c_1, c_2 \in \mathbb{R}$.

If f is continuous at $x = 0$, then $c_1 + c_2 =$ _____. (answer in integer)

gatecse-2026-set1 calculus continuity numerical-answers one-mark

Answer key

5.1.9 Continuity: GATE CSE 2026 | Set 2 | Question: 54

Consider a function $f : (0, 1) \to \{0, 1\}$ defined as follows.
For a real number $r \in (0, 1)$, $f(r) = 1$ if the second digit after the decimal point in r is one of the four digits $2, 3, 6$ and 7. Otherwise, $f(r)$ is equal to 0.

The number of points in $(0, 1)$ at which f is discontinuous is _______. (answer in integer)

gatecse-2026-set2 calculus continuity numerical-answers two-marks

Answer key

5.1.10 Continuity: GATE DS&AI 2024 | Question: 27

Let $f : \mathbb{R} \to \mathbb{R}$ be a function. Note: $\mathbb{R}$ denotes the set of real numbers.

$$f(x) = \begin{cases} -x, & \text{if } x \quad lt - 2 \\ ax^2 + bx + c, & \text{if } x \in [-2, 2] \\ x, & \text{if } x > 2 \end{cases}$$

Which ONE of the following choices gives the values of a, b, c that make the function f continuous and differentiable?

A. $a = \frac{1}{4}, b = 0, c = 1$
B. $a = \frac{1}{2}, b = 0, c = 0$
C. $a = 0, b = 0, c = 0$
D. $a = 1, b = 1, c = -4$

gate-ds-ai-2024 calculus continuity differentiation two-marks

Answer key

5.1.11 Continuity: GATE2010 ME

The function $y = |2 - 3x|$

A. **is continuous** $\forall x \in R$ and differentiable $\forall x \in R$
B. **is continuous** $\forall x \in R$ and differentiable $\forall x \in R$ except at $x = \frac{3}{2}$
C. **is continuous** $\forall x \in R$ and differentiable $\forall x \in R$ except at $x = \frac{2}{3}$
D. **is continuous** $\forall x \in R$ except $x = 3$ and differentiable $\forall x \in R$

calculus gate2010me engineering-mathematics continuity

Answer key

5.2 Definite Integral (4)

✍ **Practice Test:** Test 1 (12Q)

5.2.1 Definite Integral: GATE CSE 2023 | Question: 21

The value of the definite integral

$$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1}\left(4x^2y - z^3\right)\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$

is __________. (Rounded off to the nearest integer)

gatecse-2023 calculus definite-integral numerical-answers one-mark

Answer key☟

5.2.2 Definite Integral: GATE CSE 2024 | Set 2 | Question: 6

Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2-x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x)dx$?

A. 0 B. 1 C. 2 D. -1

gatecse-2024-set2 calculus definite-integral one-mark

Answer key☟

5.2.3 Definite Integral: GATE CSE 2025 | Set 2 | Question: 2

The value of x such that $x > 1$, satisfying the equation $\int_1^x t\ln t\,dt = \frac{1}{4}$ is

A. $\sqrt{e}$ B. e C. e^2 D. $e-1$

gatecse2025-set2 calculus definite-integral one-mark

Answer key☟

5.2.4 Definite Integral: GATE CSE 2026 | Set 2 | Question: 17

For a real number a, let $I(a) = \int_{-1}^{1}\left(3x^2 - ax + 1\right)dx$. Which of the following statements is/are true?

A. The value of $I(a)$ is independent of the value of a
B. The value of $I(a)$ can vary with the value of a
C. There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a positive real number
D. There exists $a \in (-\infty, +\infty)$ such that $I(a)$ is a negative real number

gatecse-2026-set2 calculus definite-integral multiple-selects one-mark

Answer key☟

5.3 Differentiation (11)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (10Q)

5.3.1 Differentiation: GATE CSE 1996 | Question: 1.6

The formula used to compute an approximation for the second derivative of a function f at a point x_0 is

A. $\dfrac{f(x_0+h)+f(x_0-h)}{2}$

B. $\dfrac{f(x_0+h)-f(x_0-h)}{2h}$

C. $\dfrac{f(x_0+h)+2f(x_0)+f(x_0-h)}{h^2}$

D. $\dfrac{f(x_0+h)-2f(x_0)+f(x_0-h)}{h^2}$

gate1996 calculus differentiation normal

Answer key

5.3.2 Differentiation: GATE CSE 2014 | Set 1 | Question: 46

The function $f(x) = x \sin x$ satisfies the following equation:

$$f''(x) + f(x) + t \cos x = 0$$

The value of t is______.

gatecse-2014-set1 calculus easy numerical-answers differentiation

Answer key

5.3.3 Differentiation: GATE CSE 2014 | Set 1 | Question: 6

Let the function

$$f(\theta) = \begin{vmatrix} \sin\theta & \cos\theta & \tan\theta \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) & \tan(\frac{\pi}{6}) \\ \sin(\frac{\pi}{3}) & \cos(\frac{\pi}{3}) & \tan(\frac{\pi}{3}) \end{vmatrix}$$

where

$\theta \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ and $f'(\theta)$ denote the derivative of f with respect to θ. Which of the following statements is/are **TRUE**?

I. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) = 0$
II. There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) \neq 0$

A. I only
B. II only
C. Both I and II
D. Neither I nor II

gatecse-2014-set1 calculus differentiation normal

Answer key

5.3.4 Differentiation: GATE CSE 2016 | Set 2 | Question: 02

Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is __________.

gatecse-2016-set2 calculus normal numerical-answers differentiation

Answer key

5.3.5 Differentiation: GATE CSE 2017 | Set 2 | Question: 10

If $f(x) = R\ \sin(\frac{\pi x}{2}) + S, f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2R}{\pi}$, then the constants R and S are

A. $\frac{2}{\pi}$ and $\frac{16}{\pi}$
B. $\frac{2}{\pi}$ and 0
C. $\frac{4}{\pi}$ and 0
D. $\frac{4}{\pi}$ and $\frac{16}{\pi}$

gatecse-2017-set2 engineering-mathematics calculus differentiation

Answer key

5.3.6 Differentiation: GATE CSE 2024 | Set 1 | Question: 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\left\{x, x^3\right\}, x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

A. $\{-1, 1, 2\}$
B. $\{-2, -1, 1\}$
C. $\{0, 1\}$
D. $\{-1, 0, 1\}$

gatecse-2024-set1 calculus differentiation one-mark

Answer key

5.3.7 Differentiation: GATE CSE 2025 | Set 1 | Question: 21

Consider the given function $f(x)$.

$$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$

If the function is differentiable everywhere, the value of b must be ________. (rounded off to one decimal place)

gatecse2025-set1 calculus differentiation numerical-answers one-mark

Answer key

5.3.8 Differentiation: GATE CSE 2026 | Set 1 | Question: 36

Let $f : \mathbb{R} \to \mathbb{R}$ be defined as follows:

$$f(x) = \left(\frac{|x|}{2} - x\right)\left(x - \frac{|x|}{2}\right)$$

Which of the following statements is/are true?

A. f has a local maximum
B. f has a local minimum
C. f' is continuous over $\mathbb{R}$
D. f' is not differentiable over $\mathbb{R}$

gatecse-2026-set1 two-marks differentiation calculus multiple-selects

Answer key

5.3.9 Differentiation: GATE DA 2025 | Question: 14

Consider two functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to (1, \infty)$. Both functions are differentiable at a point c. Which of the following functions is/are ALWAYS differentiable at c ? The symbol $\cdot$ denotes product and the symbol $\circ$ denotes composition of functions.

A. $f \pm g$
B. $f \cdot g$
C. $\frac{f}{g}$
D. $f \circ g + g \circ f$

gateda-2025 calculus differentiation limits multiple-selects one-mark

Answer key

5.3.10 Differentiation: GATE DA 2025 | Question: 4

Let $f(x) = \frac{e^x - e^{-x}}{2}, x \in \mathbb{R}$. Let $f^{(k)}(a)$ denote the k^{th} derivative of f evaluated at a. What is the value of $f^{(10)}(0)$? (Note: ! denotes factorial)

A. 0
B. 1
C. $\frac{1}{10!}$
D. $\frac{2}{10!}$

gateda-2025 calculus differentiation one-mark

Answer key

5.3.11 Differentiation: GATE DA 2025 | Question: 41

Let $f : \mathbb{R} \to \mathbb{R}$ be a twice-differentiable function and suppose its second derivative satisfies $f''(x) > 0$ for all $x \in \mathbb{R}$. Which of the following statements is/are ALWAYS correct?

A. f has a local minima
B. There does not exist x and $y, x \neq y$, such that $f'(x) = f'(y) = 0$
C. f has at most one global minimum
D. f has at most one local minimum

gateda-2025 calculus maxima-minima differentiation multiple-selects two-marks

Answer key

5.4 Integration (11)

✍ **Practice Test:** Test 1 (14Q)

5.4.1 Integration: GATE CSE 1993 | Question: 02.6

The value of the double integral $\int_0^1 \int_0^{\frac{1}{x}} \frac{x}{1+y^2} dx dy$ is__________.

gate1993 calculus integration normal fill-in-the-blanks

Answer key☟

5.4.2 Integration: GATE CSE 1998 | Question: 8

a. Find the points of local maxima and minima, if any, of the following function defined in $0 \leq x \leq 6$.

$$x^3 - 6x^2 + 9x + 15$$

b. Integrate

$$\int_{-\pi}^{\pi} x \cos x dx$$

gate1998 calculus maxima-minima integration normal descriptive

Answer key☟

5.4.3 Integration: GATE CSE 2000 | Question: 2.3

Let $S = \sum_{i=3}^{100} i \log_2 i$, and $T = \int_2^{100} x \log_2 x dx$.
Which of the following statements is true?

A. $S > T$
B. $S = T$
C. $S < T$ and $2S > T$
D. $2S \leq T$

gatecse-2000 calculus integration normal

Answer key☟

5.4.4 Integration: GATE CSE 2009 | Question: 25

$\int_0^{\pi/4} (1 - \tan x)/(1 + \tan x) \, dx$

A. 0
B. 1
C. $\ln 2$
D. $1/2 \ln 2$

gatecse-2009 calculus integration normal

Answer key☟

5.4.5 Integration: GATE CSE 2011 | Question: 31

Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int\limits_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$?

A. 0
B. 2
C. $-i$
D. i

gatecse-2011 calculus integration normal

Answer key☟

5.4.6 Integration: GATE CSE 2014 | Set 3 | Question: 47

The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x \, dx$$

A. -2π B. π C. $-\pi$ D. 2π

gatecse-2014-set3 calculus limits integration normal

Answer key☟

5.4.7 Integration: GATE CSE 2014 | Set 3 | Question: 6

If $\int\limits_0^{2\pi} |x \ \sin x| dx = k\pi$, then the value of k is equal to _______.

gatecse-2014-set3 calculus integration limits numerical-answers easy

Answer key☟

5.4.8 Integration: GATE CSE 2015 | Set 1 | Question: 44

Compute the value of:

$$\int_{\frac{1}{\pi}}^{\frac{2}{\pi}} \frac{\cos(1/x)}{x^2} dx$$

gatecse-2015-set1 calculus integration normal numerical-answers

Answer key☟

5.4.9 Integration: GATE CSE 2015 | Set 3 | Question: 45

If for non-zero x, $af(x) + bf(\frac{1}{x}) = \frac{1}{x} - 25$ where $a \neq b$ then $\int\limits_1^2 f(x)dx$ is

A. $\frac{1}{a^2-b^2}\left[a(\ln 2 - 25) + \frac{47b}{2}\right]$

B. $\frac{1}{a^2-b^2}\left[a(2\ln 2 - 25) - \frac{47b}{2}\right]$

C. $\frac{1}{a^2-b^2}\left[a(2\ln 2 - 25) + \frac{47b}{2}\right]$

D. $\frac{1}{a^2-b^2}\left[a(\ln 2 - 25) - \frac{47b}{2}\right]$

gatecse-2015-set3 calculus integration normal

Answer key☟

5.4.10 Integration: GATE CSE 2018 | Question: 16

The value of $\int_0^{\pi/4} x \cos(x^2) dx$ correct to three decimal places (assuming that $\pi = 3.14$) is ____

gatecse-2018 calculus integration normal numerical-answers one-mark

Answer key☟

5.4.11 Integration: GATE IT 2005 | Question: 35

What is the value of $\int_0^{2\pi} (x - \pi)^2 (\sin x) dx$

A. -1 B. 0 C. 1 D. π

gateit-2005 calculus integration normal

Answer key☟

5.5 Limits (15)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (8Q)

5.5.1 Limits: GATE CSE 1993 | Question: 02.1

$\lim_{x \to 0} \frac{x(e^x - 1) + 2(\cos x - 1)}{x(1 - \cos x)}$ is ________

gate1993 limits calculus normal fill-in-the-blanks

Answer key

5.5.2 Limits: GATE CSE 1995 | Question: 7(B)

Compute without using power series expansion $\lim_{x \to 0} \frac{\sin x}{x}$.

gate1995 calculus limits numerical-answers

Answer key

5.5.3 Limits: GATE CSE 2008 | Question: 1

$\lim_{x \to \infty} \frac{x - \sin x}{x + \cos x}$ equals

A. 1

B. -1

C. ∞

D. $-\infty$

gatecse-2008 calculus limits easy

Answer key

5.5.4 Limits: GATE CSE 2010 | Question: 5

What is the value of $\lim_{n \to \infty} \left(1 - \frac{1}{n}\right)^{2n}$?

A. 0

B. e^{-2}

C. $e^{-1/2}$

D. 1

gatecse-2010 calculus limits normal

Answer key

5.5.5 Limits: GATE CSE 2015 | Set 1 | Question: 4

$\lim_{x \to \infty} x^{\frac{1}{x}}$ is

A. ∞

B. 0

C. 1

D. Not defined

gatecse-2015-set1 calculus limits normal

Answer key

5.5.6 Limits: GATE CSE 2015 | Set 3 | Question: 9

The value of $\lim_{x \to \infty} (1 + x^2)^{e^{-x}}$ is

A. 0

B. $\frac{1}{2}$

C. 1

D. ∞

gatecse-2015-set3 calculus limits normal

Answer key

5.5.7 Limits: GATE CSE 2016 | Set 1 | Question: 3

$$\lim_{x \to 4} \frac{\sin(x - 4)}{x - 4} = \_\_\_\_\_\_\_\_$$

gatecse-2016-set1 calculus limits easy numerical-answers

Answer key

5.5.8 Limits: GATE CSE 2017 | Set 1 | Question: 28

The value of $\lim_{x \to 1} \frac{x^7 - 2x^5 + 1}{x^3 - 3x^2 + 2}$

A. is 0
B. is -1
C. is 1
D. does not exist

gatecse-2017-set1 calculus limits normal

Answer key

5.5.9 Limits: GATE CSE 2019 | Question: 13

Compute $\lim_{x \to 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$

A. 1
B. $53/12$
C. $108/7$
D. Limit does not exist

gatecse-2019 engineering-mathematics calculus limits one-mark

Answer key

5.5.10 Limits: GATE CSE 2021 | Set 1 | Question: 20

Consider the following expression.

$$\lim_{x \to -3} \frac{\sqrt{2x + 22} - 4}{x + 3}$$

The value of the above expression (rounded to 2 decimal places) is ___________.

gatecse-2021-set1 calculus limits numerical-answers one-mark

Answer key

5.5.11 Limits: GATE CSE 2022 | Question: 24

The value of the following limit is ________________.

$$\lim_{x \to 0^+} \frac{\sqrt{x}}{1 - e^{2\sqrt{x}}}$$

gatecse-2022 numerical-answers calculus limits one-mark

Answer key

5.5.12 Limits: GATE DA 2025 | Question: 22

$$\lim_{t \to +\infty} \sqrt{t^2 + t} - t =$$

(Round off to one decimal place)

gateda-2025 calculus limits numerical-answers one-mark

Answer key

5.5.13 Limits: GATE DA 2025 | Question: 49

Let $f : \mathbb{R} \to \mathbb{R}$ be such that $|f(x) - f(y)| \leq (x - y)^2$ for all $x, y \in \mathbb{R}$. Then $f(1) - f(0) =$ __________
(Answer in integer)

gateda-2025 calculus limits numerical-answers two-marks

Answer key

5.5.14 Limits: GATE DS&AI 2024 | Question: 50

Evaluate the following limit:

$$\lim_{x \to 0} \frac{\ln\left(\left(x^2+1\right)\cos x\right)}{x^2} =$$

gate-ds-ai-2024 calculus numerical-answers limits engineering-mathematics two-marks

Answer key

5.5.15 Limits: GATE Data Science and Artificial Intelligence 2024 | Sample Paper | Question: 5

$\lim_{x \to 2} \frac{\sqrt{x}-\sqrt{2}}{x-2}$

A. 0

B. $\sqrt{2}$

C. $\frac{1}{2\sqrt{2}}$

D. $\frac{1}{\sqrt{2}}$

gateda-sample-paper-2024 calculus limits

Answer key

5.6 Maxima Minima (14)

✍ **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (6Q)

5.6.1 Maxima Minima: GATE CSE 1987 | Question: 1-xxvi

If $f(x_i).\, f(x_{i+1}) < 0$ then

A. There must be a root of $f(x)$ between x_i and x_{i+1}
B. There need not be a root of $f(x)$ between x_i and x_{i+1}
C. There fourth derivative of $f(x)$ with respect to x vanishes at x_i
D. The fourth derivative of $f(x)$ with respect to x vanishes at x_{i+1}

gate1987 calculus maxima-minima

Answer key

5.6.2 Maxima Minima: GATE CSE 1995 | Question: 1.21

In the interval $[0, \pi]$ the equation $x = \cos x$ has

A. No solution
B. Exactly one solution
C. Exactly two solutions
D. An infinite number of solutions

gate1995 calculus normal maxima-minima

Answer key

5.6.3 Maxima Minima: GATE CSE 1995 | Question: 25a

Find the minimum value of $3 - 4x + 2x^2$.

gate1995 calculus maxima-minima easy descriptive

Answer key

5.6.4 Maxima Minima: GATE CSE 1997 | Question: 4.1

What is the maximum value of the function $f(x) = 2x^2 - 2x + 6$ in the interval $[0, 2]$?

A. 6
B. 10
C. 12
D. 5.5

gate1997 calculus maxima-minima normal

Answer key

5.6.5 Maxima Minima: GATE CSE 2008 | Question: 25

A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is

A. 0
B. 1
C. 2
D. 3

gatecse-2008 calculus maxima-minima easy

Answer key

5.6.6 Maxima Minima: GATE CSE 2012 | Question: 9

Consider the function $f(x) = \sin(x)$ in the interval $x = \left[\frac{\pi}{4}, \frac{7\pi}{4}\right]$. The number and location(s) of the local minima of this function are

A. One, at $\dfrac{\pi}{2}$
B. One, at $\dfrac{3\pi}{2}$
C. Two, at $\dfrac{\pi}{2}$ and $\dfrac{3\pi}{2}$
D. Two, at $\dfrac{\pi}{4}$ and $\dfrac{3\pi}{2}$

gatecse-2012 calculus maxima-minima normal

Answer key

5.6.7 Maxima Minima: GATE CSE 2015 | Set 2 | GA | Question: 3

Consider a function $f(x) = 1 - |x| \text{ on } -1 \leq x \leq 1.$ The value of x at which the function attains a maximum, and the maximum value of the function are:

A. $0, -1$
B. $-1, 0$
C. $0, 1$
D. $-1, 2$

gatecse-2015-set2 set-theory&algebra functions normal maxima-minima

Answer key

5.6.8 Maxima Minima: GATE CSE 2020 | Question: 1

Consider the functions

I. e^{-x}
II. $x^2 - \sin x$
III. $\sqrt{x^3 + 1}$

Which of the above functions is/are increasing everywhere in $[0, 1]$?

A. III only
B. II only
C. II and III only
D. I and III only

gatecse-2020 engineering-mathematics calculus maxima-minima one-mark

Answer key

5.6.9 Maxima Minima: GATE CSE 2023 | Question: 18

Let

$$f(x) = x^3 + 15x^2 - 33x - 36$$

be a real-valued function.

Which of the following statements is/are TRUE?

A. $f(x)$ does not have a local maximum.
B. $f(x)$ has a local maximum.

C. $f(x)$ does not have a local minimum.

D. $f(x)$ has a local minimum.

gatecse-2023 calculus maxima-minima multiple-selects one-mark

Answer key☟

5.6.10 Maxima Minima: GATE DA 2025 | Question: 39

Consider the function $f(x)=\dfrac{x^3}{3}+\dfrac{7}{2}x^2+10x+\dfrac{133}{2}, x\in[-8,0]$. Which of the following statements is/are correct?

A. The maximum value of f is attained at $x=-5$
B. The minimum value of f is attained at $x=-2$
C. The maximum value of f is $\dfrac{133}{2}$
D. The minimum value of the derivative of f is attained at $x=-\dfrac{7}{2}$

gateda-2025 calculus maxima-minima multiple-selects two-marks

Answer key☟

5.6.11 Maxima Minima: GATE DA 2026 | Question: 17

Let $f(x)=x^3-3x^2+2$ be a function defined on $(-1,3]$.
Which of the following statements is/are correct?

A. $f(x)$ has exactly two roots in $[-0.9,0]$.

B. $f(x)$ has a minimum at 2 only.

C. $f(x)$ has maximum at 0 only.

D. $f(x)$ has a root at 1.

gateda-2026 calculus maxima-minima multiple-selects one-mark

Answer key☟

5.6.12 Maxima Minima: GATE DS&AI 2024 | Question: 40

Consider the function $f:\mathbb{R}\to\mathbb{R}$ where $\mathbb{R}$ is the set of all real numbers.

$$f(x)=\frac{x^4}{4}-\frac{2x^3}{3}-\frac{3x^2}{2}+1$$

Which of the following statements is/are TRUE?

A. $x=0$ is a local maximum of f

B. $x=3$ is a local minimum of f

C. $x=-1$ is a local maximum of f

D. $x=0$ is a local minimum of f

gate-ds-ai-2024 calculus maxima-minima multiple-selects two-marks

Answer key☟

5.6.13 Maxima Minima: GATE DS&AI 2024 | Question: 5

For any twice differentiable function $f:\mathbb{R}\to\mathbb{R}$, if at some $x^*\in\mathbb{R}, f'(x^*)=0$ and $f''(x^*)>0$, then the function f necessarily has a ________ at $x=x^*$.

Note: $\mathbb{R}$ denotes the set of real numbers.

A. local minimum

B. global minimum

C. local maximum

D. global maximum

gate-ds-ai-2024 calculus maxima-minima one-mark

Answer key☟

5.6.14 Maxima Minima: GATE IT 2008 | Question: 31

If $f(x)$ is defined as follows, what is the minimum value of $f(x)$ for $x \in (0, 2]$?

$$f(x) = \begin{cases} \frac{25}{8x} & \text{when } x \leq \frac{3}{2} \\ x + \frac{1}{x} & \text{otherwise} \end{cases}$$

A. 2
B. $2\frac{1}{12}$
C. $2\frac{1}{6}$
D. $2\frac{1}{2}$

gateit-2008 calculus maxima-minima normal

Answer key

5.7 Polynomials (2)

5.7.1 Polynomials: GATE CSE 1987 | Question: 1-xxii

The equation $7x^7 + 14x^6 + 12x^5 + 3x^4 + 12x^3 + 10x^2 + 5x + 7 = 0$ has

A. All complex roots
B. At least one real root
C. Four pairs of imaginary roots
D. None of the above

gate1987 calculus polynomials

Answer key

5.7.2 Polynomials: GATE CSE 1995 | Question: 2.8

If the cube roots of unity are $1, \omega$ and ω^2, then the roots of the following equation are

$$(x-1)^3 + 8 = 0$$

A. $-1, 1+2\omega, 1+2\omega^2$
B. $1, 1-2\omega, 1-2\omega^2$
C. $-1, 1-2\omega, 1-2\omega^2$
D. $-1, 1+2\omega, -1+2\omega^2$

gate1995 calculus normal polynomials

Answer key

5.8 Summation (1)

5.8.1 Summation: GATE DA 2026 | Question: 25

The value of $\sum_{i=0}^{\infty} \sum_{j=1}^{\infty} 2^{-i} 3^{-j}$ is _____. (Answer in integer)

gateda-2026 calculus numerical-answers one-mark summation

Answer key

Answer Keys

5.1.1	N/A	5.1.2	B	5.1.3	A	5.1.4	A	5.1.5	A
5.1.6	C	5.1.7	19 : 19	5.1.8	03 : 03	5.1.9	40:40	5.1.10	A
5.1.11	C	5.2.1	0	5.2.2	B	5.2.3	A	5.2.4	A;C
5.3.1	D	5.3.2	-2	5.3.3	C	5.3.4	9	5.3.5	C
5.3.6	D	5.3.7	-2.1:-1.9	5.3.8	A;C;D	5.3.9	A;B;C	5.3.10	A
5.3.11	B;C;D	5.4.1	N/A	5.4.2	N/A	5.4.3	A	5.4.4	D
5.4.5	D	5.4.6	A	5.4.7	4	5.4.8	-1	5.4.9	A
5.4.10	0.288 : 0.289	5.4.11	B	5.5.1	1	5.5.2	1	5.5.3	A
5.5.4	B	5.5.5	C	5.5.6	C	5.5.7	1	5.5.8	C
5.5.9	C	5.5.10	0.25 : 0.25	5.5.11	-0.5	5.5.12	0.5:0.5	5.5.13	0:0